

BPM 1

MSK: Lecture 25

PHYSIOLOGY OF CARDIAC AND SMOOTH MUSCLES

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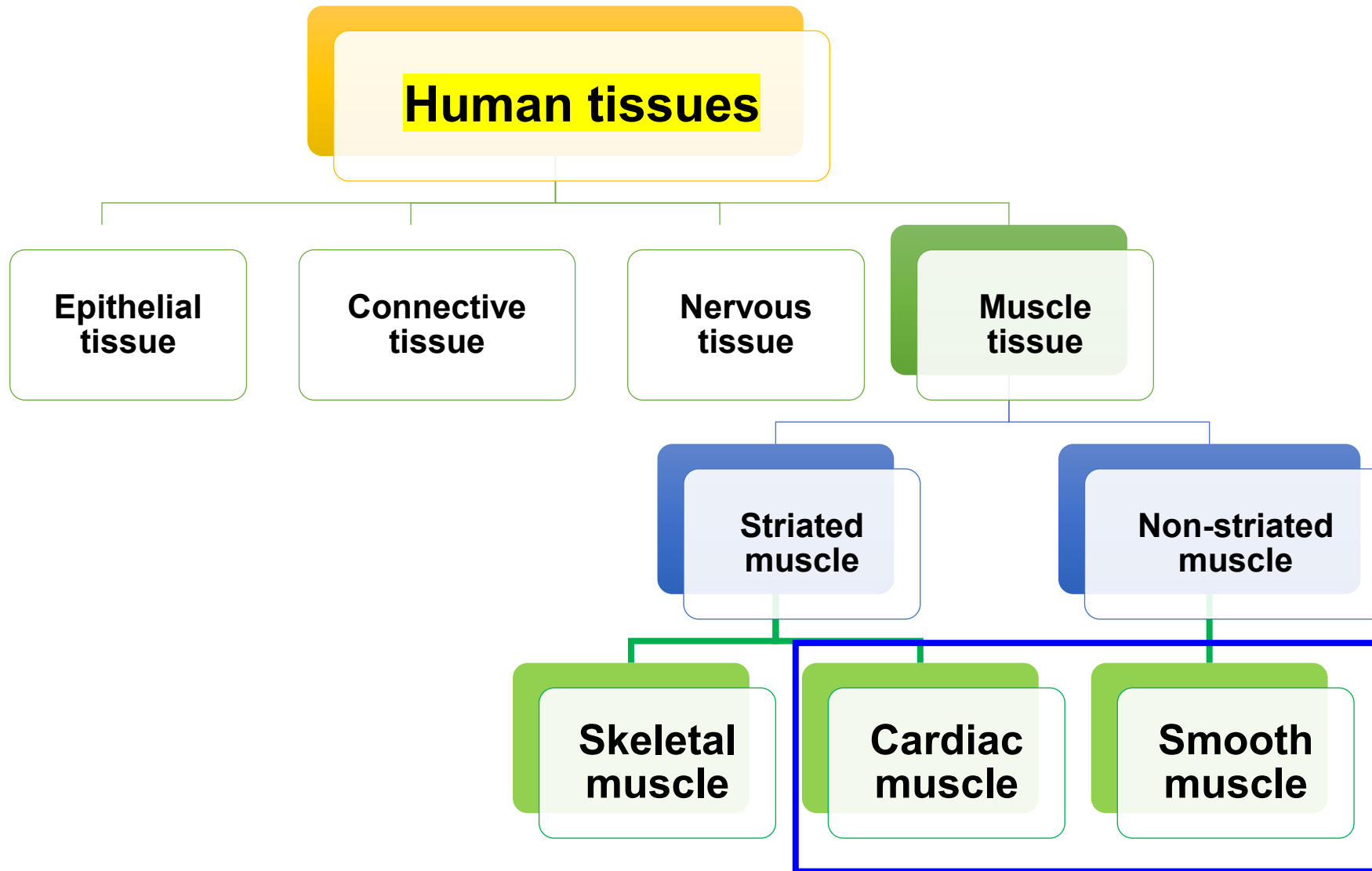
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Recommended Reading

Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell. Chapter 13: Mechanics of cardiac pump. Read under headings, “Introduction” and “Excitation-contraction coupling in cardiac muscle”.

Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell. Chapter 8: Skeletal and smooth muscle. Read under headings, “Smooth Muscle” and “Activation-contraction coupling in smooth muscle”.

Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know



CARDIAC MUSCLE

LEARNING OBJECTIVES

SOM.MK.II.BPM1.2.MSK.2.PHYS.0067 Contrast the duration of the action potential and the refractory period in a cardiac muscle, a skeletal muscle, and a nerve. Sketch the temporal relationship between an action potential in a cardiac muscle cell and the resulting contraction (twitch) of that cell. On the basis of that graph, explain why cardiac muscle cannot remain in a state of sustained (tetanic) contraction.

SOM.MK.II.BPM1.2.MSK.2.PHYS.0068 List the steps in excitation-contraction coupling in the cardiac myocytes from the initiation of an action potential to the relaxation of the cell.

SOM.MK.II.BPM1.2.MSK.2.PHYS.0069 Provide specific details about the special role of Ca^{2+} in the control of contraction and relaxation of cardiac muscle.

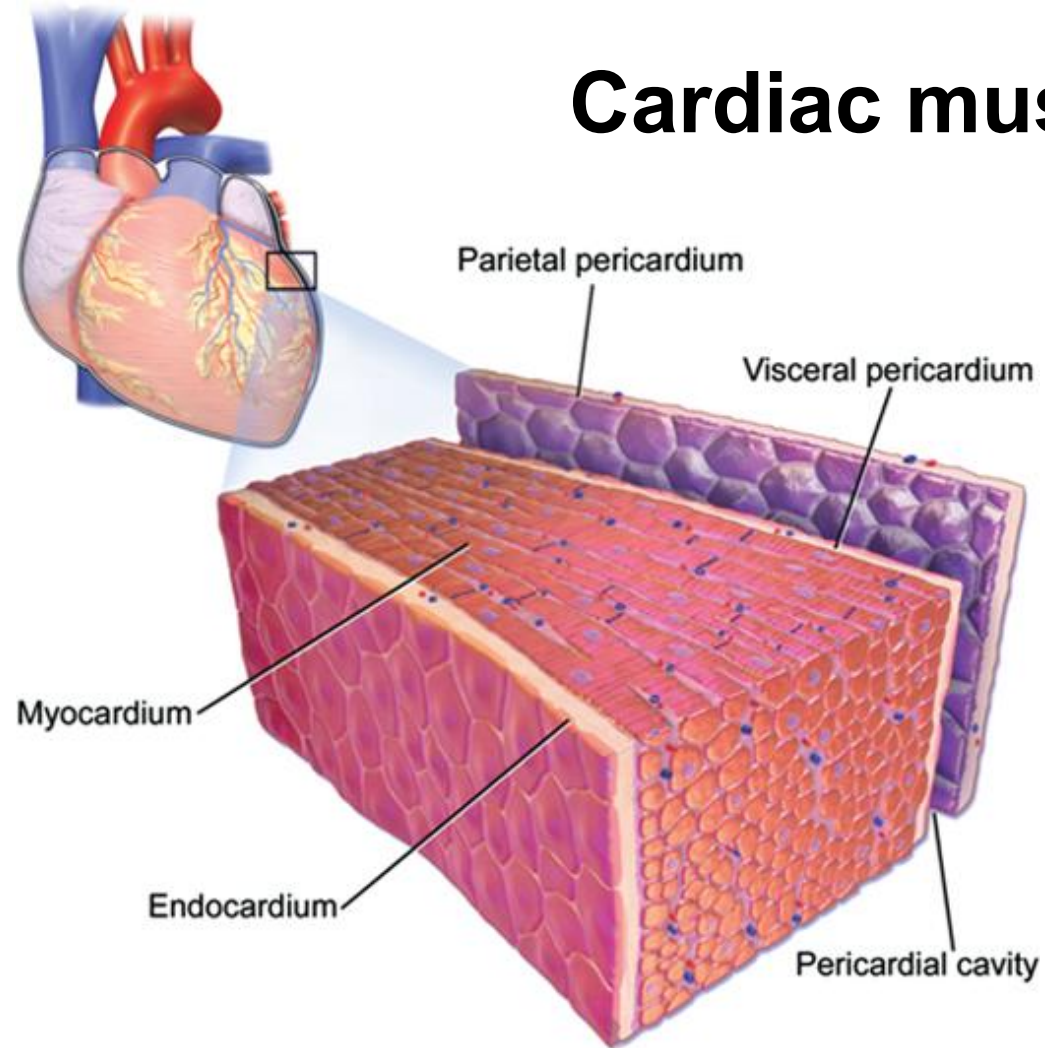
SOM.MK.II.BPM1.2.MSK.2.PHYS.0070 Compare cardiac and skeletal muscle with respect to: cell size, electrical connections between cells, and arrangement of myofilaments. Based on ion permeability and electrical resistance describe role of gap junctions in creating a functional syncytium.

SOM.MK.II.BPM1.2.MSK.2.PHYS.0071 Identify the role of extracellular calcium in cardiac muscle contraction. Identify other sources of calcium that mediate excitation-contraction coupling and describe how intracellular calcium concentration modulates the strength of cardiac muscle contraction (Inotropism).

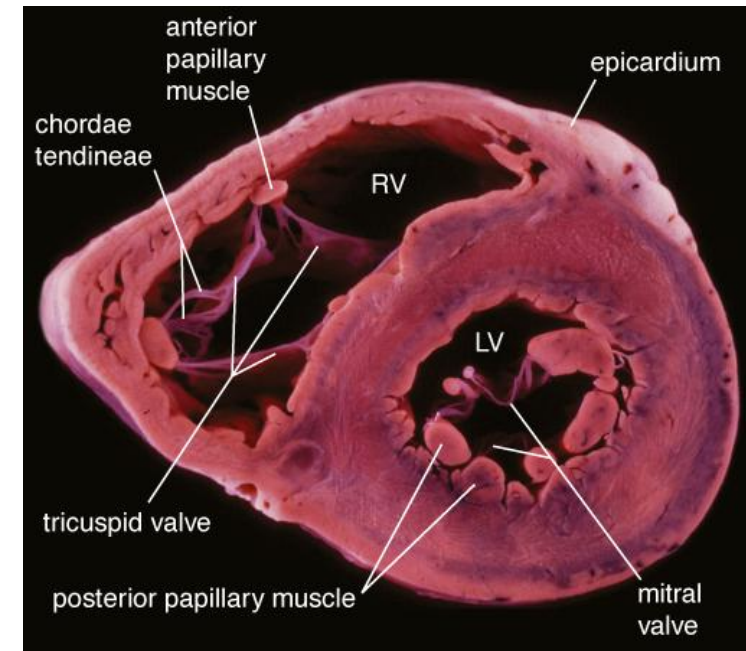
Lecture outline: Cardiac muscle

- ❖ Cardiac muscle structure and action potential
- ❖ Functional syncytium of the cardiac muscle
- ❖ Excitation-contraction coupling and role of calcium in contraction/relaxation
- ❖ Role of extracellular calcium and inotropism in heart function

Cardiac muscle



Source: Valentin Fuster, Robert A. Harrington, Jagat Narula, Zubin J. Eapen: Hurst's The Heart, Fourteenth Edition: www.accessmedicine.com Copyright © McGraw-Hill Education. All rights reserved.

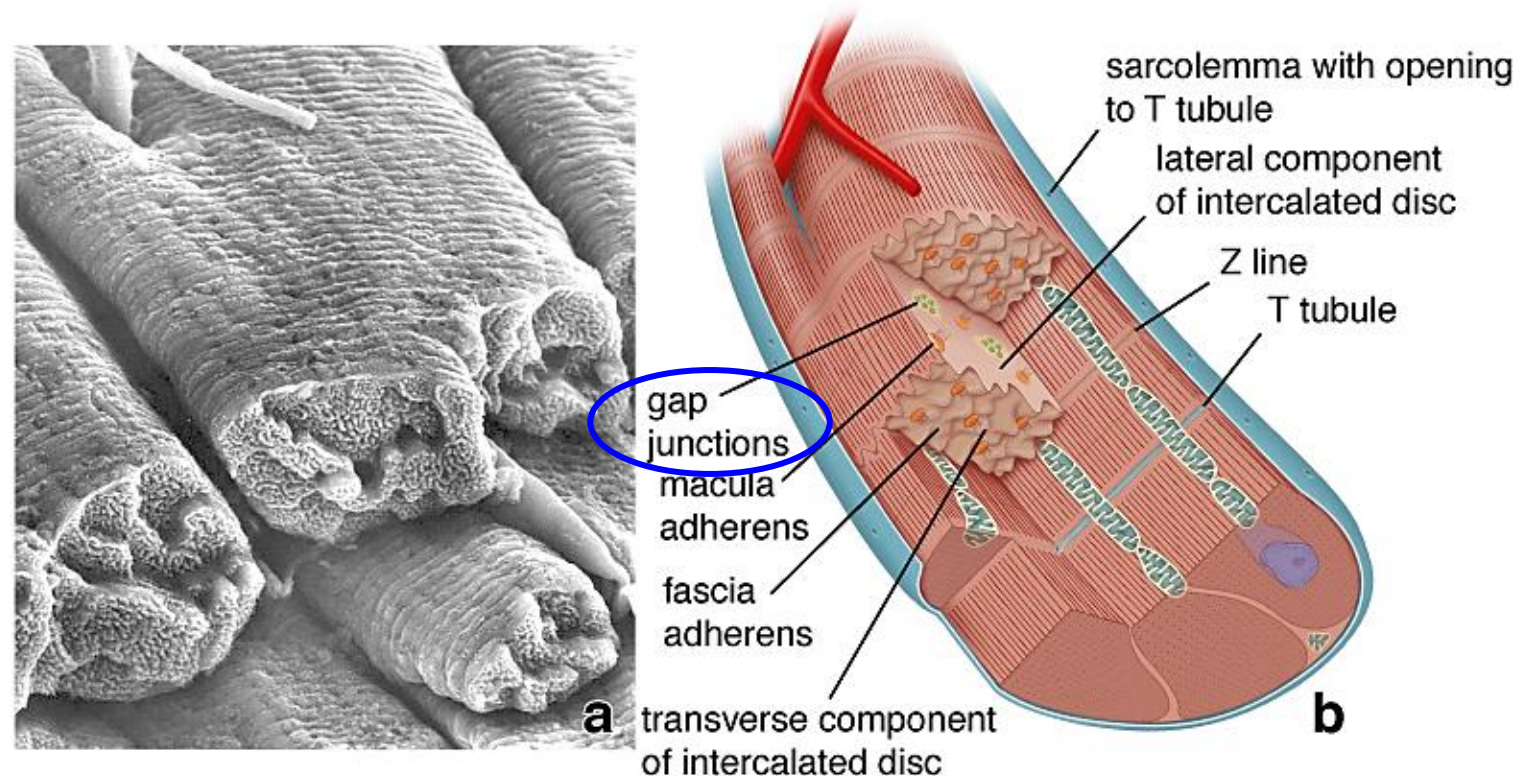


Histology: A Text and Atlas: With Correlated Cell and Molecular Biology, 8e, 2020

Myocardium forms the walls of atria and ventricles that pump blood throughout the vascular system

Cardiac muscle cells: Ultrastructure

- Cylindric, branched cells
- Size:
 - Diameter 25 μm
 - Length 100 μm
- 1-3 nuclei, centrally located
- **Intercalated discs** connect cells



Intercalated discs: Functional syncytium

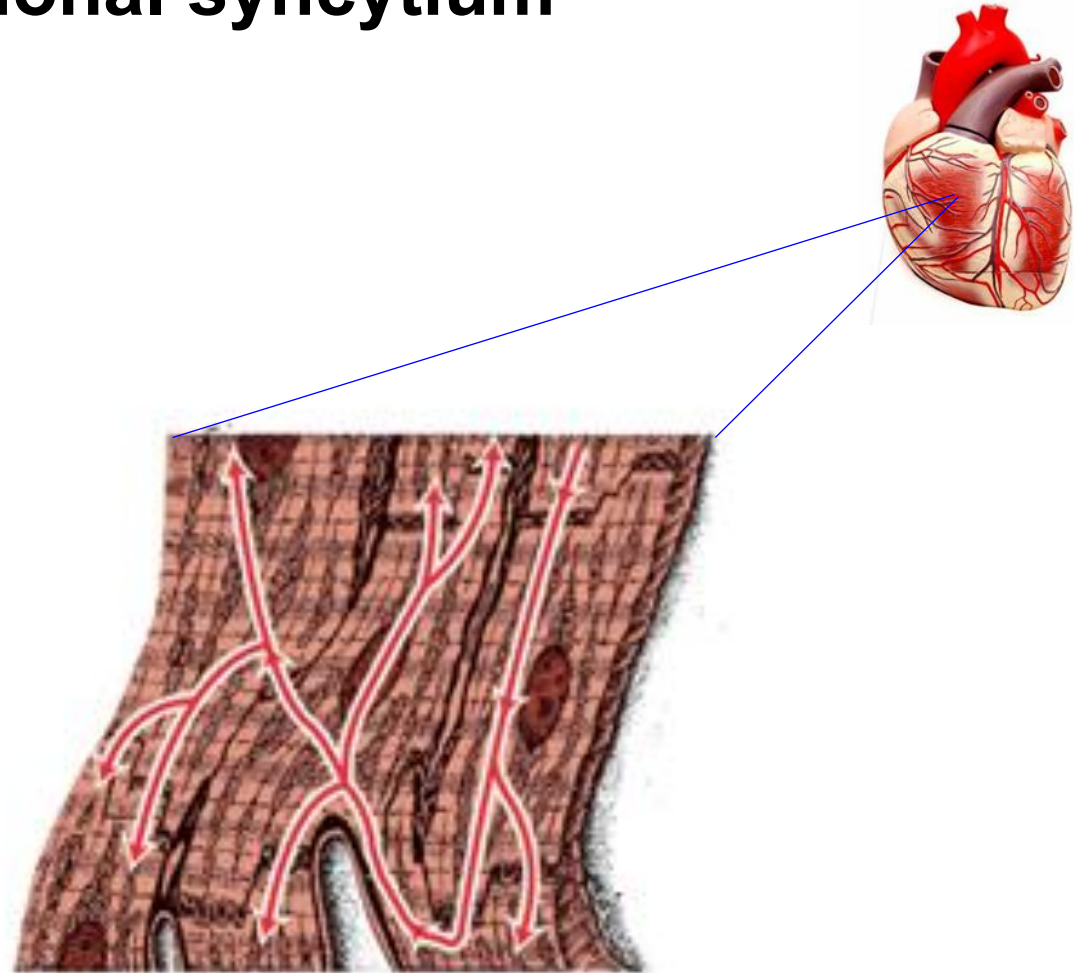
- Transverse components:

- Fascia adherens

- Desmosomes

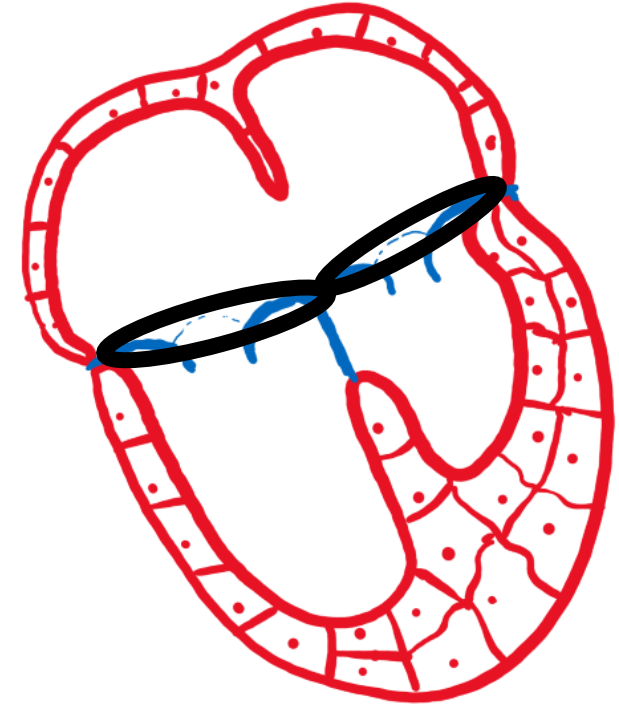
- Lateral components

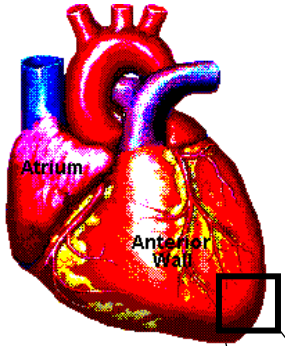
- **Gap junctions**



Intercalated discs: Functional syncytium

- GAP junctions provide **least electrical resistance**
- All cells in atria (upper chambers) or ventricles (lower chambers) are electrically interconnected
- **No spatial summation** possible in heart
- FUNCTIONAL SYNCYTIUM:
 - **Atrial syncytium** (both upper chambers contract together)
 - **Ventricular syncytium** (both lower chambers contract together)





Cardiac Action Potentials: Types

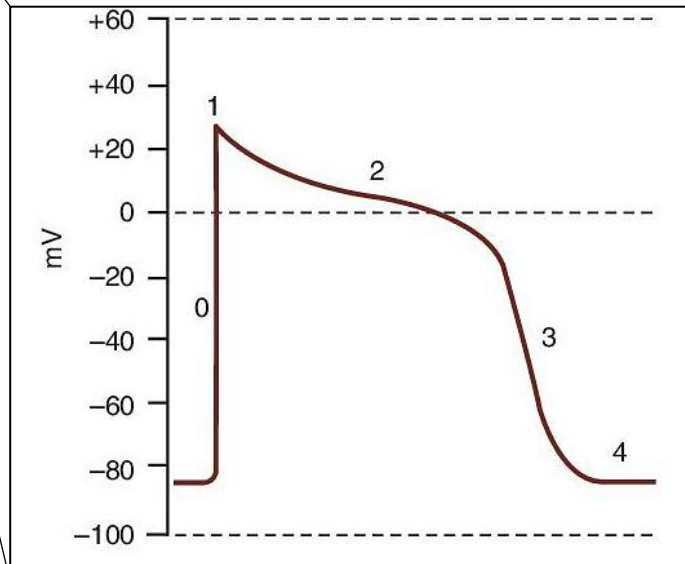
Two broad types of cardiac action potentials:

Fast response type:

- Ventricular muscle
- Atrial muscle and
- Purkinje fibers.

Slow response type:

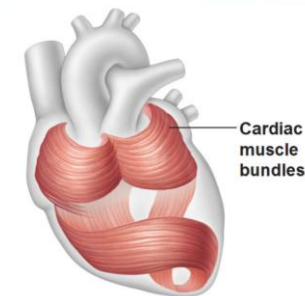
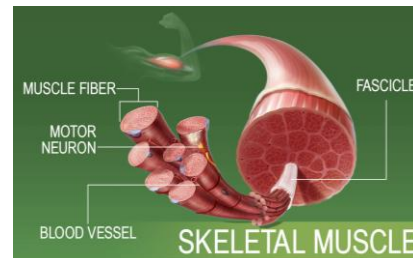
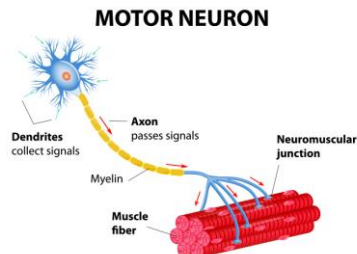
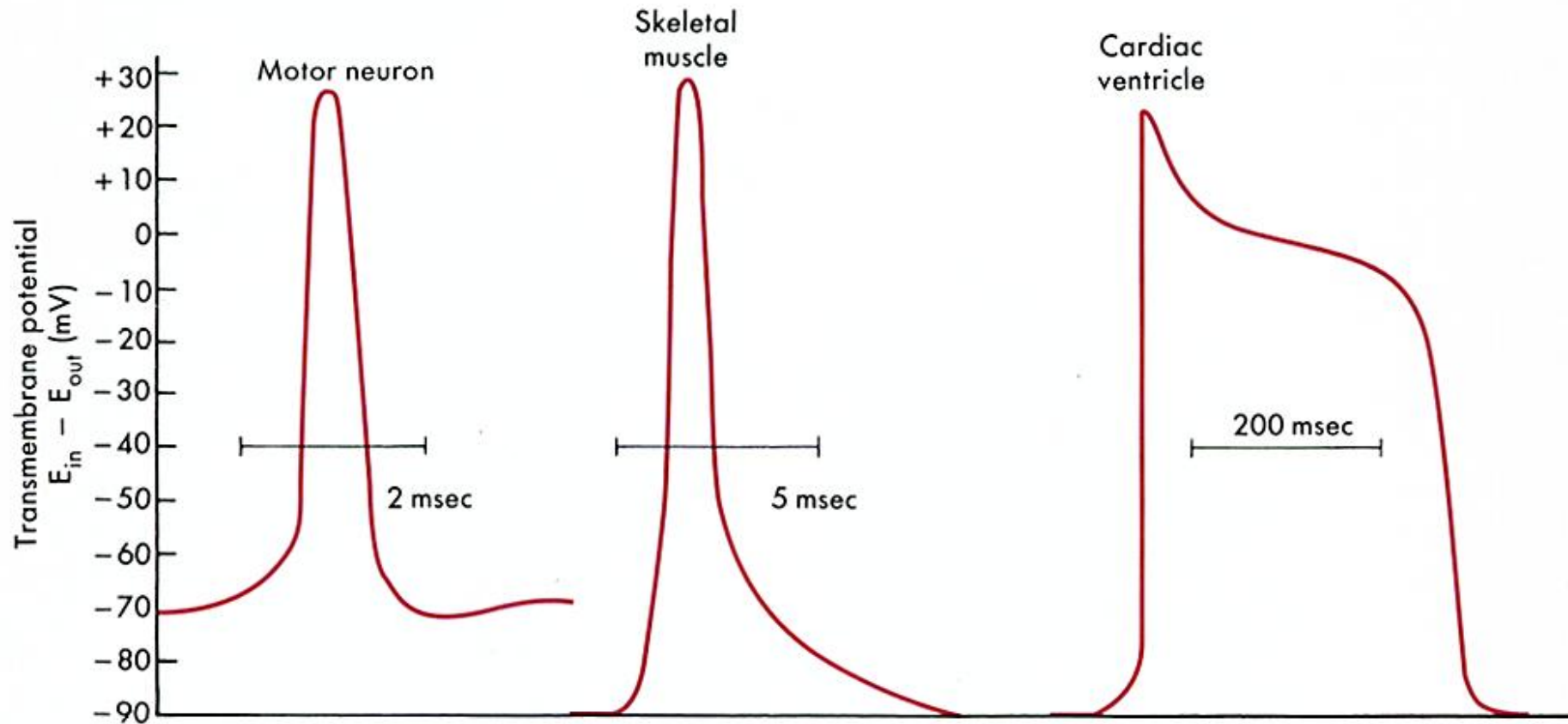
- Sinoatrial (SA) node and
- Atrioventricular node (AV node)



Numbers in graph indicate phases of the AP

Heart generates its own action potential (Automaticity)

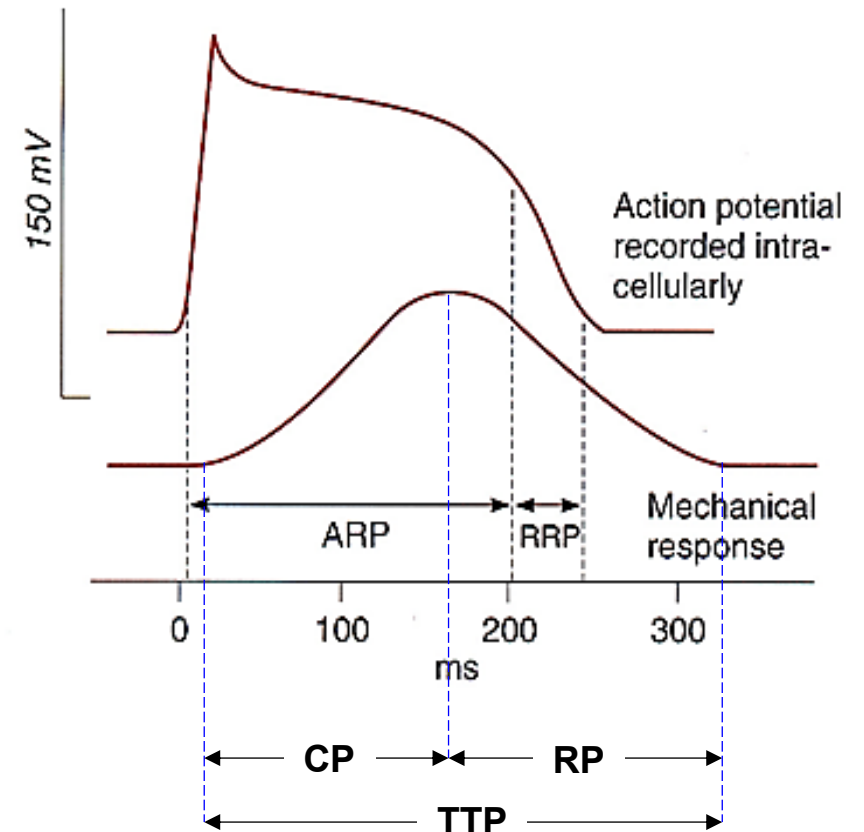
Cardiac Action Potential has a long duration



Importance of Long duration of cardiac AP

- Absolute refractory period (ARB) is also long.
- **No temporal summation** possible
- Tetanic contraction is **NOT possible** in the myocardium
- Protects heart from too frequent excitation
- Helps heart to work as an **effective pump**

Imagine if the ventricles remain in sustained contraction.....



ARP: Absolute refractory period

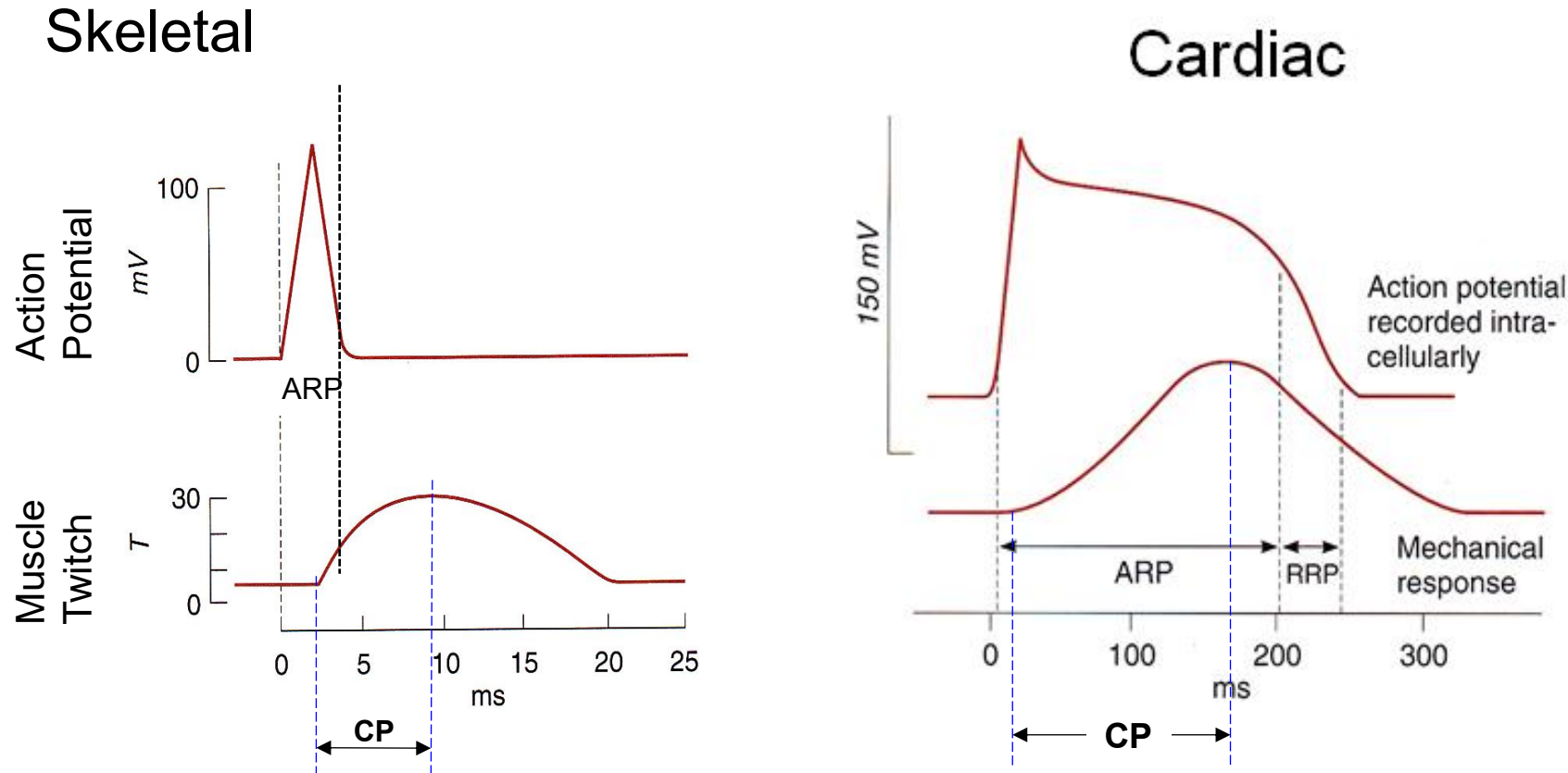
RRP: Relative refractory period

CP: Contraction period

RP: Relaxation period

TTP: Total twitch period

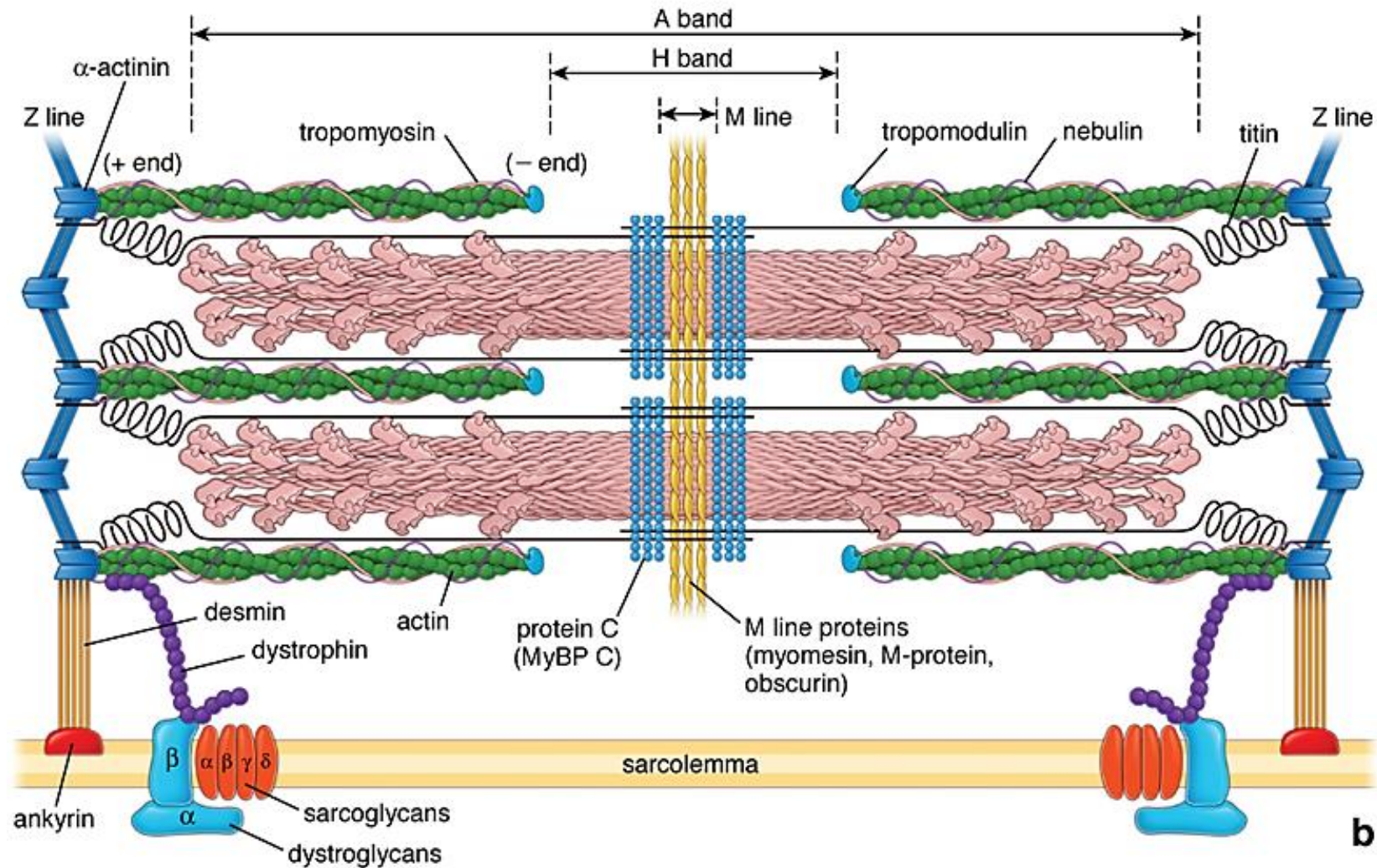
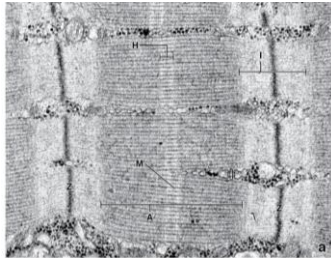
Comparison of electrical (Action potential) and mechanical (contraction) events in skeletal and cardiac muscles



Refractory period **exceeds contraction period (CP)** in cardiac muscle

Excitation-contraction coupling in cardiac muscle

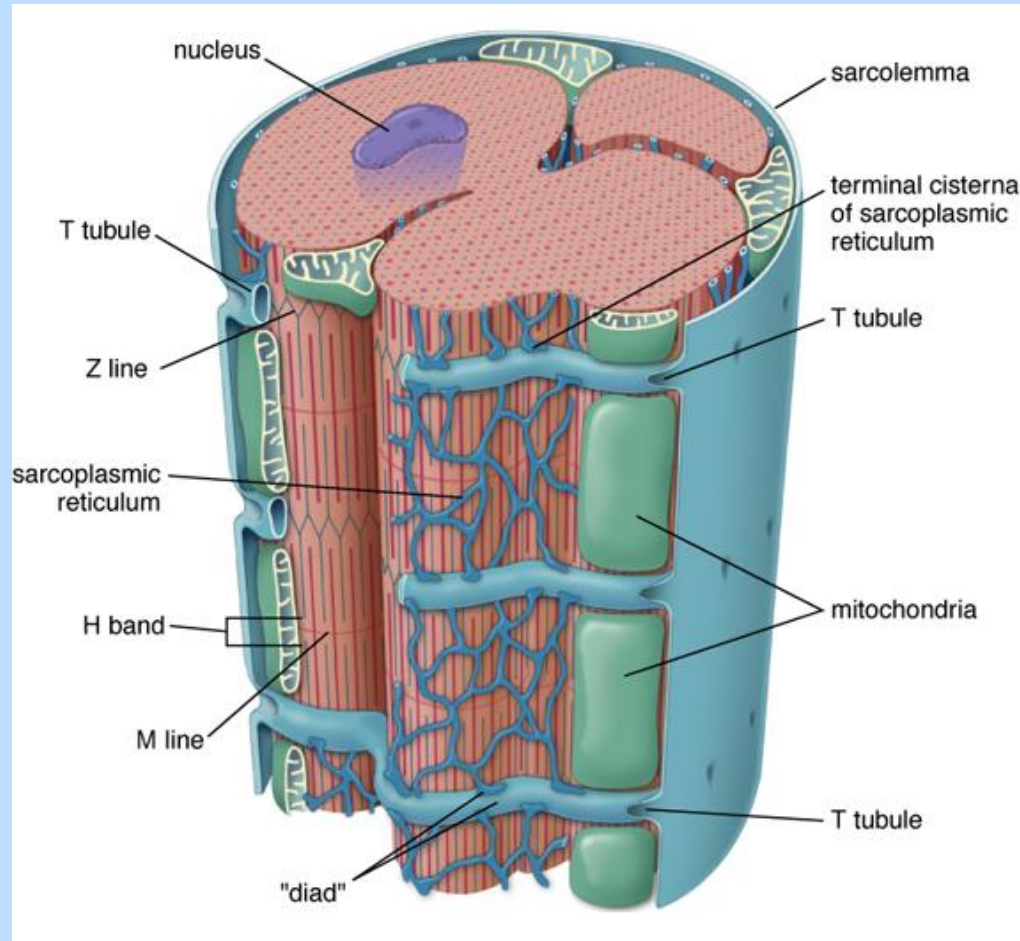
SARCOMERE: contractile unit in striated muscle



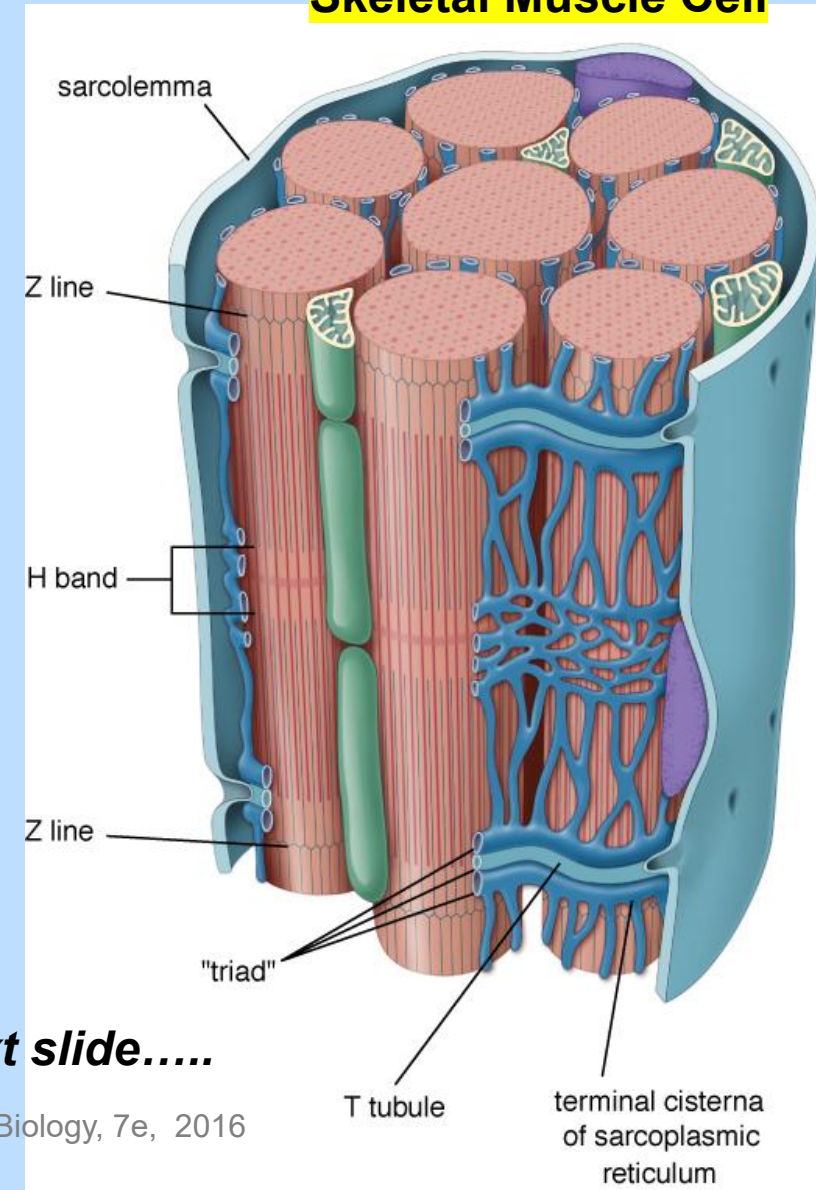
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Comparison of T tubules and Sarcoplasmic Reticulum in striated muscles

Cardiac Muscle Cell



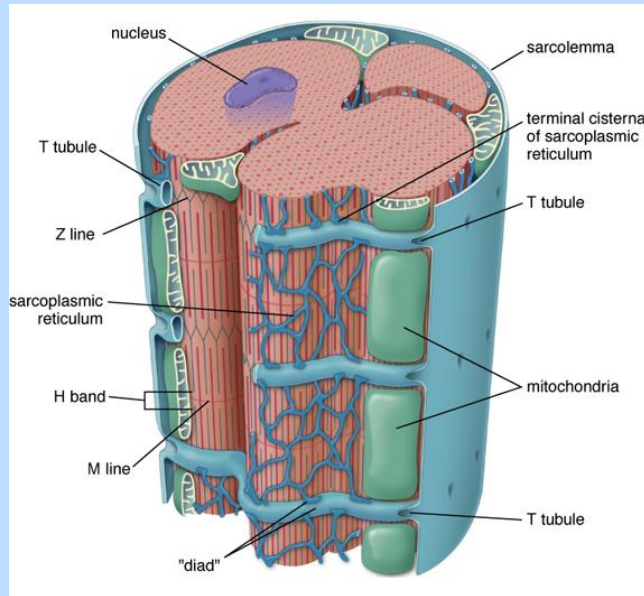
Skeletal Muscle Cell



Explanation in the next slide.....

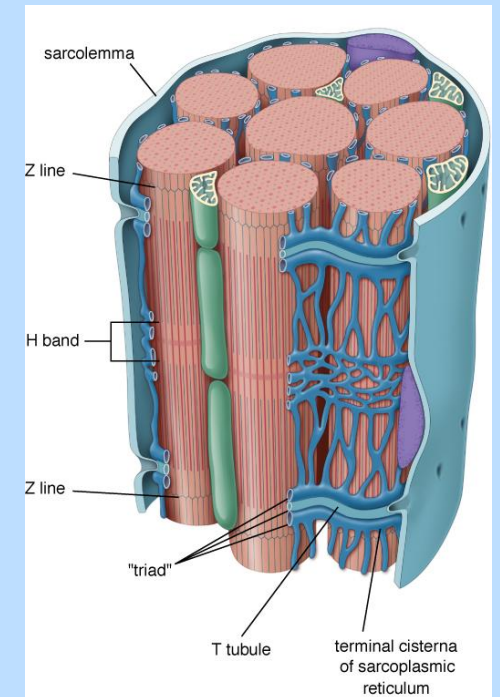
Histology: A Text and Atlas with Correlated Cell and Molecular Biology, 7e, 2016

Comparison of T tubules and Sarcoplasmic Reticulum in striated muscles



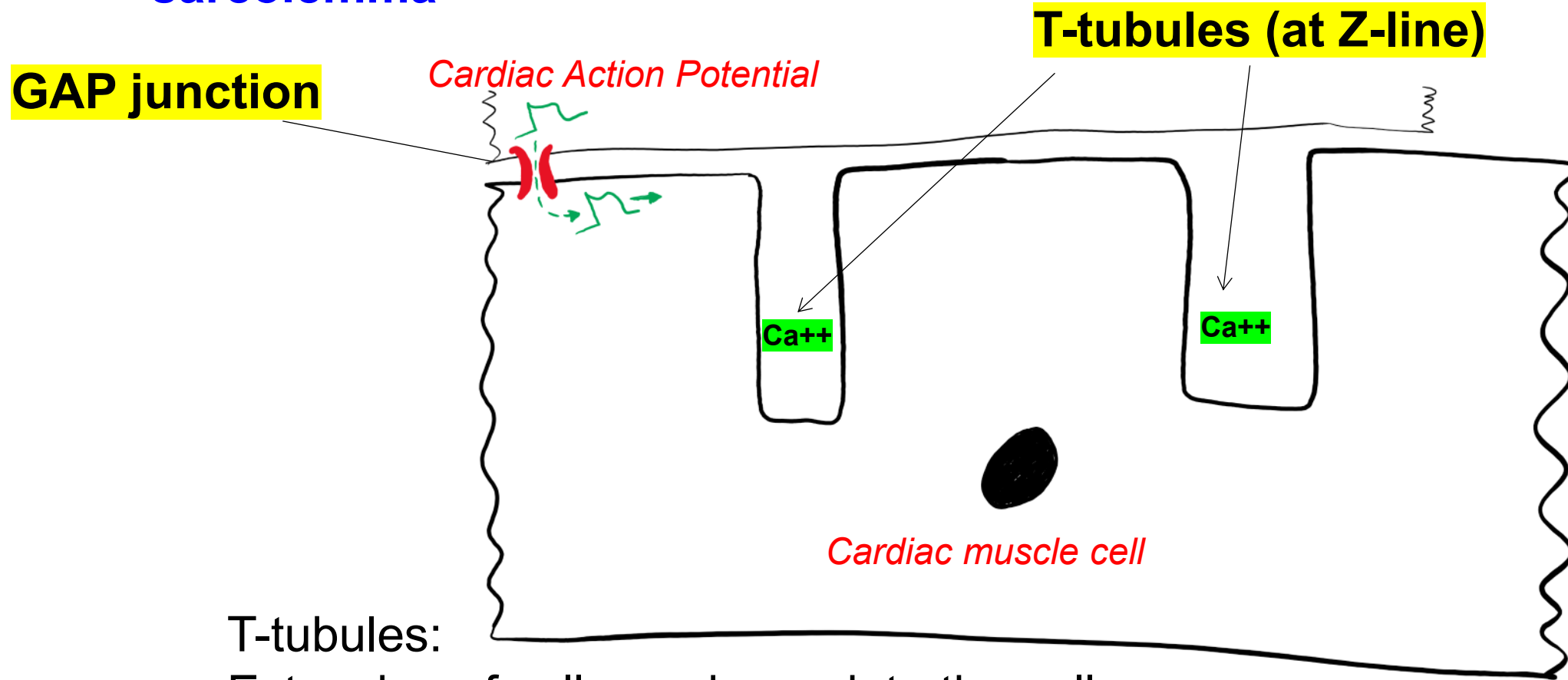
The T tubules of cardiac muscle are much larger than the T tubules of skeletal muscle and carry an investment of external lamina material into the cell. They are **located at the level of the Z line**. The portion of the sarcoplasmic reticulum adjacent to the T tubule is not in the form of an expanded cisterna but rather is organized as a “**diad**” anastomosing network.

In skeletal muscle fibers, two transverse (T) tubules supply a sarcomere. Each T tubule is located at an **A–I band junction** and is formed as an invagination of the sarcolemma of skeletal muscle. It is associated with two terminal cisternae of the sarcoplasmic reticulum that surrounds each myofibril, one cisterna on either side of the T tubule. The triple structure is called a **triad**.



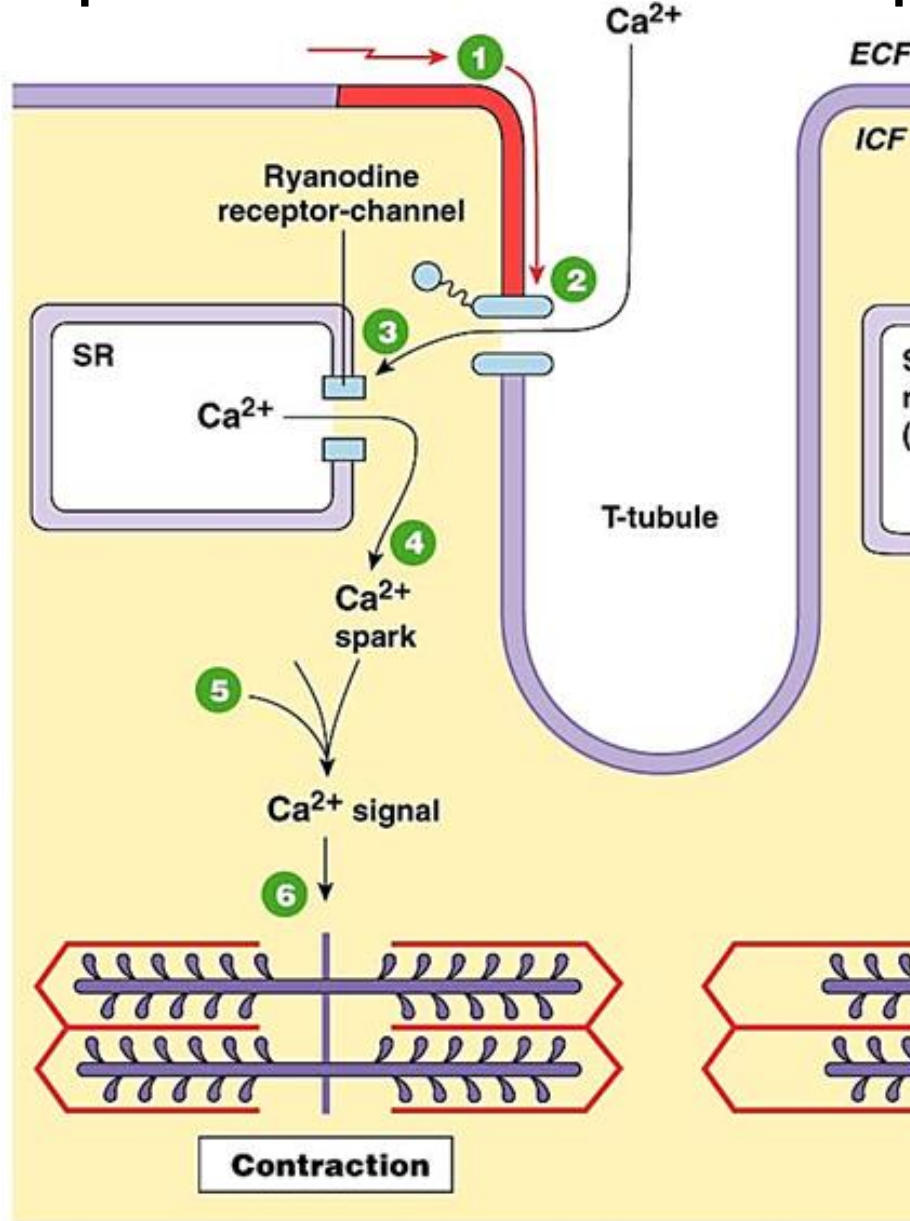
Steps of Excitation-contraction coupling in cardiac muscle

Step 1: Arrival of an Action Potential and depolarization of the sarcolemma



T-tubules:
Extension of cell membrane into the cell,
Contains extracellular fluid & Ca⁺⁺

Steps of Excitation-contraction coupling in cardiac muscle



Steps 2 to 6:

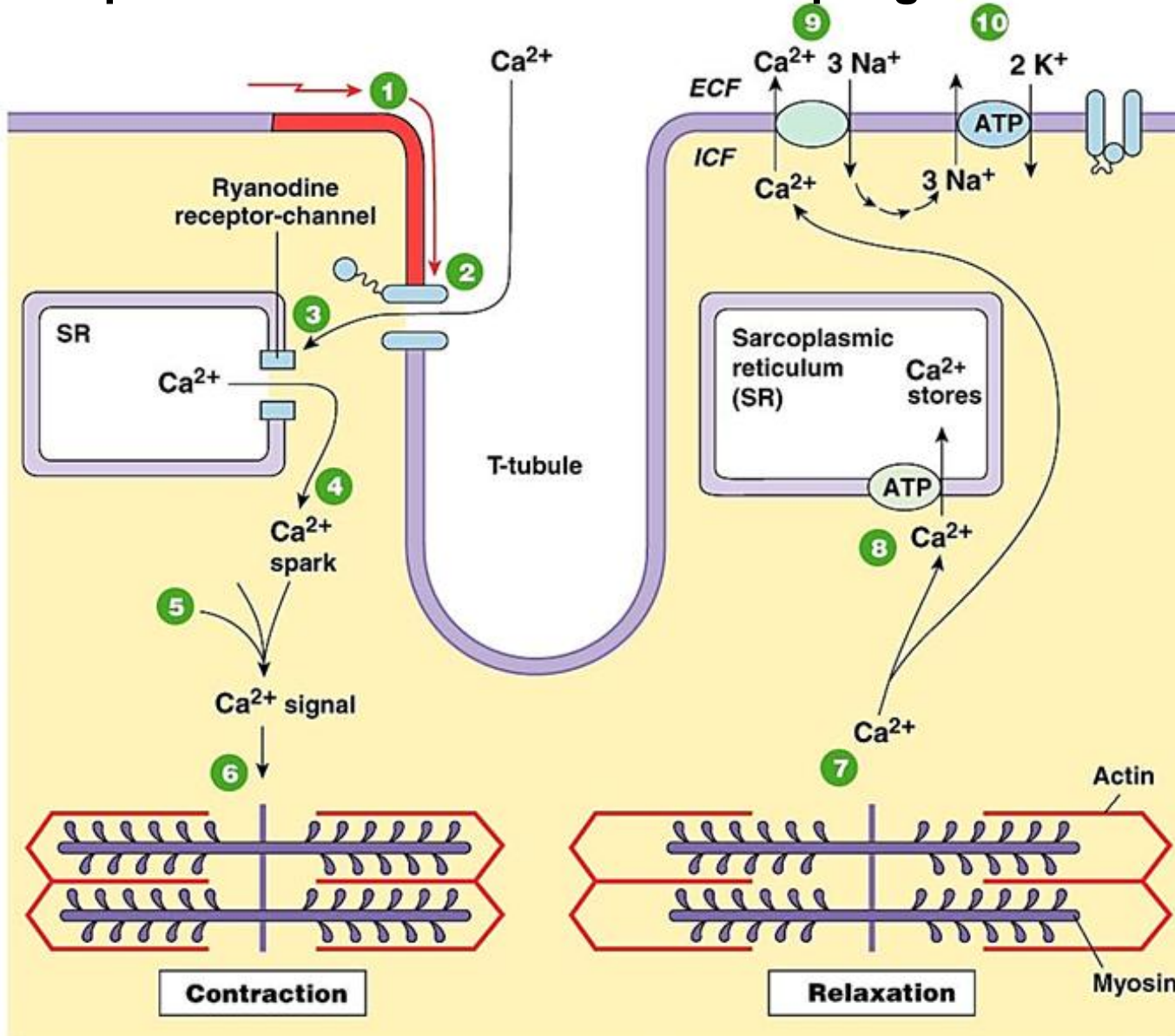
- 2 Voltage-gated Ca^{2+} channels open. Ca^{2+} enters cell. (L-type Ca^{2+} channels or Dihydropyridine channels)
- 3 Ca^{2+} induces Ca^{2+} release through ryanodine receptor-channels (RyR).
- 4 Local release causes Ca^{2+} spark. **Ca^{2+} -induced Ca^{2+} release**
- 5 Summed Ca^{2+} sparks create a Ca^{2+} signal.
- 6 Ca^{2+} ions bind to troponin to initiate contraction.

Entry of ECF Ca^{2+} is **ESSENTIAL** in cardiac muscle contraction

Stored calcium also matters!

Steps of Excitation-contraction coupling in cardiac muscle

Steps 7 to 10: Relaxation

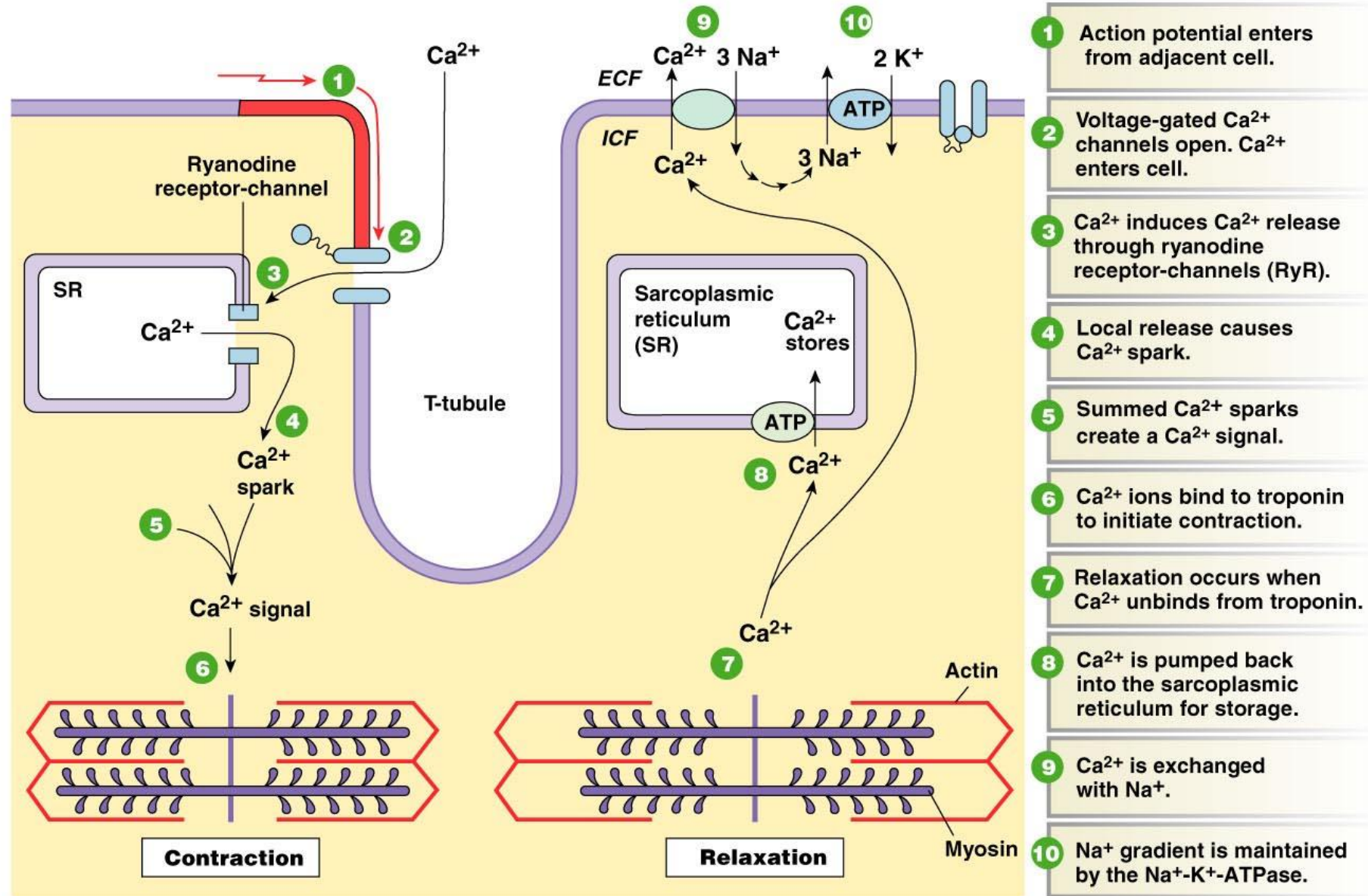


- 7 Relaxation occurs when Ca^{2+} unbinds from troponin.
- 8 Ca^{2+} is pumped back into the sarcoplasmic reticulum for storage. **(Ca++-ATPase pump – 80%)**
- 9 Ca^{2+} is exchanged with Na^{+} . **(Na-Ca²⁺ Exchange, 20%)**
- 10 Na^{+} gradient is maintained by the **Na⁺-K⁺-ATPase**.

Removal of Ca^{2+} from cytoplasm is ESSENTIAL in cardiac muscle relaxation

Rigor mortis also occurs in heart!

Summary of Excitation-Contraction coupling



EXCITATION-CONTRACTION COUPLING: Difference between cardiac and skeletal muscle

Excitation–contraction coupling in the cardiac muscle differs from that in the skeletal muscle in that it may be **modulated**, that is, different intensities of actin–myosin interaction (contraction) can result from a single action potential trigger in the cardiac muscle. The mechanism for this is largely dependent on variations in the amount of Ca^{2+} reaching the myofilaments and therefore the number of cross-bridges activated during the contraction. This ability of the cardiac muscle to vary its contractile strength—that is, to change its *contractility*—is extremely important to cardiac function.

Comparison between Striated Muscles: SKELETAL vs CARDIAC

Features	Skeletal muscle cells	Cardiac muscle cells
Nucleus of the cell	Multiple, peripheral nuclei	Central, one or two nuclei
SHAPE of the cell	Non-branched cylindric	Branched cylindric
SIZE of the cell	Diameter: 25 μm Length: almost entire length of muscle (several cm)	Diameter: 25 μm Length: 100 μm
Electrical properties	Cells electrically isolated from each other, each cell receive innervation, cannot contract without nerve stimulation	All the cells form electrical syncytium (GAP junctions), AUTOMATICITY
	Action potential has short duration (like nerve cells): 5 msec	Action potential has a long duration: 200-300 msec
Source of activator Ca^{++}	Sarcoplasmic reticulum (SR)	Extracellular Ca^{++} and SR "calcium-induced calcium release"
Arrangements of myofilaments	Actin and myosin filaments arranged in sarcomeres (contractile units)	

Heart continues to function even after denervation₃₂

Role of calcium in heart muscle performance

- Movement of ECF Ca^{2+} is essential for contraction
- Amount of Ca^{2+} released from the sarcoplasmic reticulum (SR) depends on the amount of Ca^{++} moving into the cell during the action potential
- Amount of calcium released also depends on the stored Ca^{++} in the SR.
- Force of contraction **correlates directly** with the intracellular Ca^{++} concentration

Greater the intracellular Ca^{2+} concentration → Greater the force of contraction

Molecular mechanism: Binding of calcium to TN-C is determined by the free intracellular concentration of calcium and the binding affinity of TN-C to calcium. $\uparrow$ Calcium concentration $\rightarrow \uparrow$ Calcium binding to TN-C $\rightarrow$ Greater force generated between actin and myosin.

Inotropism (contractility) and calcium

- Contractility, or inotropism, is the intrinsic ability of myocardial cells to develop force at a given muscle cell length.
- Increase in contractility: **Positive inotropic** effect;
- Decrease in contractility: **Negative inotropic** effect.
- All positive inotropic agents **elevate intracellular Calcium.**

Factors/agents that alter inotropism

Positive inotropism

- Sympathetic nerve stimulation
- Catecholamines (Epi, NE and DA)
- Exercise
- Beta receptor agonists
- Cardiac glycoside (Digoxin)
- PDE-3 inhibitor (Milrinone)

E.g., **Sympathomimetic drug** like Dobutamine which is a Beta1 receptor agonist, is useful in shock due to myocardial infarction

Negative inotropism

- Sympathetic inhibition
- Myocardial loss
- Beta receptor antagonists
- Calcium channel blockers

(E=Epinephrine; NE=Norepinephrine; DA=Dopamine)

TWO mechanisms increase Ca^{+2} in the myocardium and produce positive inotropism (stronger contraction of the ventricles)

1. Increasing Ca^{+2} influx into the myocardium

Beta1 agonists (E, NE & DA) and
PDE-3 inhibitor (Milrinone)

2. Reducing Ca^{+2} efflux from the myocardium

Cardiac glycosides:
Digoxin, Digitoxin

Conversely, agents that block **calcium influx** to the cell reduce the performance of the heart (E.g., Calcium channel blockers and beta blockers).

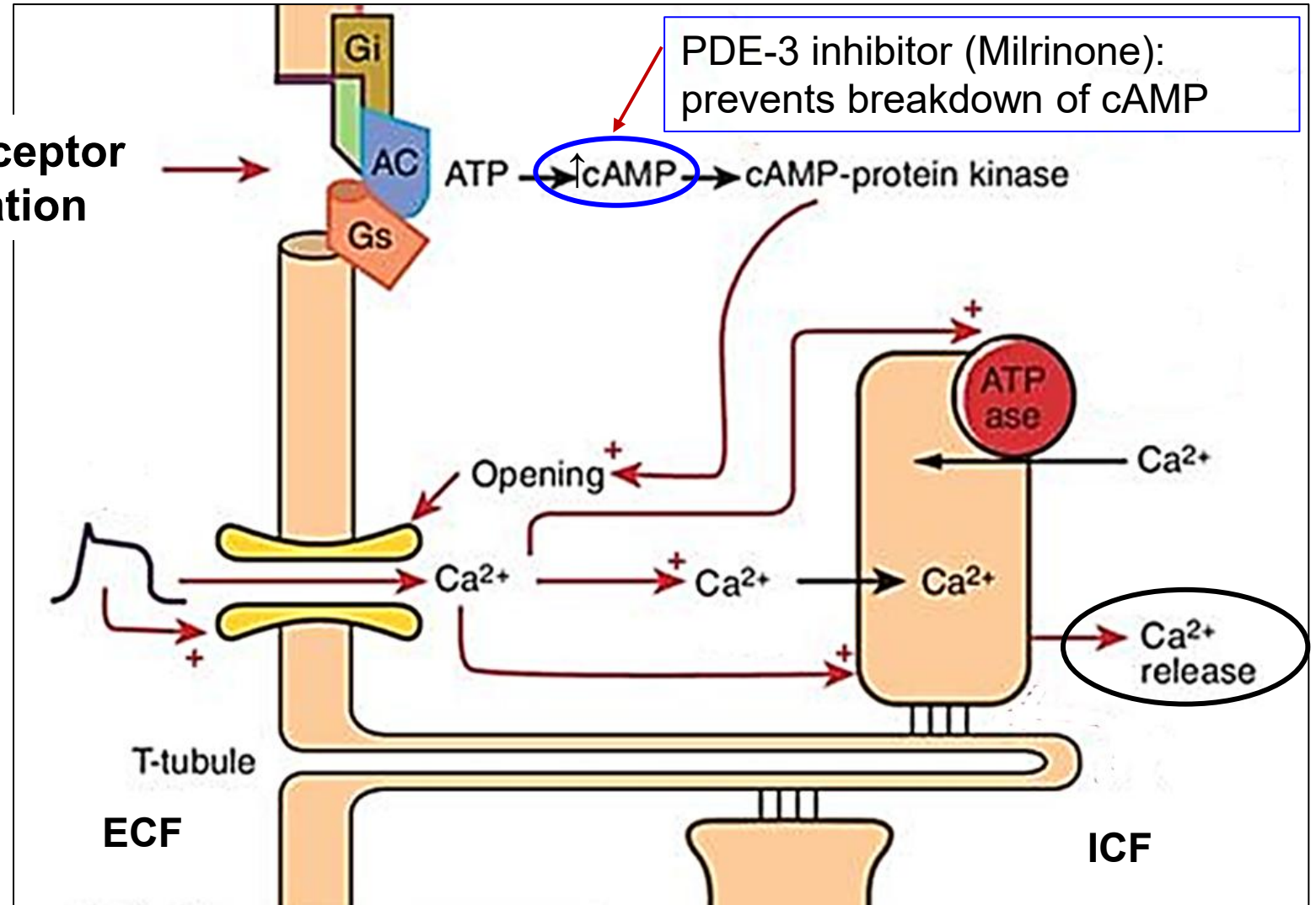
Intracellular calcium modulates cardiac muscle contraction

(PDE-3 = Phosphodiesterase-3, enzyme that breaks down cAMP)

1. Increasing Ca^{2+} entry occurs by increasing cAMP inside the myocardium

Catecholamines,
Sympathetic stimulation,
Beta-receptor agonists,
Exercise

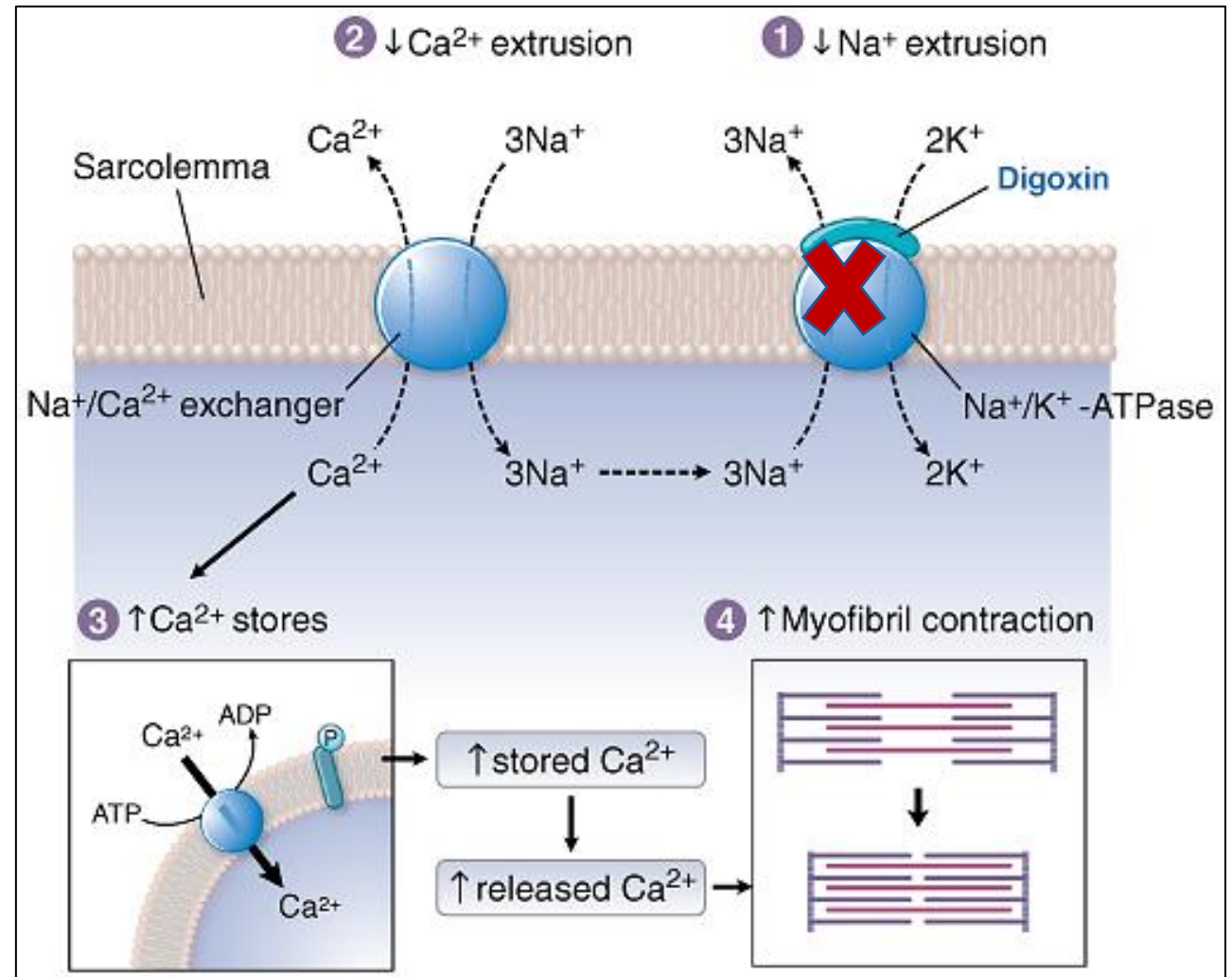
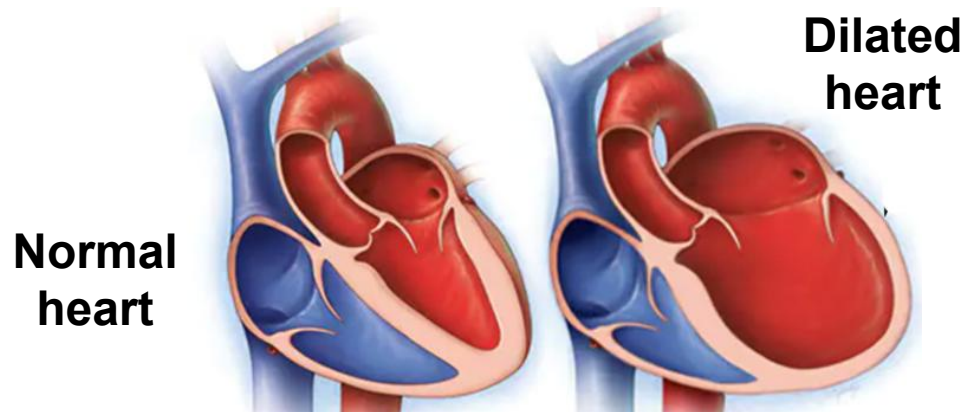
Beta1 receptor
stimulation



2. Cardiac glycosides (Digoxin, digitoxin) produce positive inotropism by reducing Ca^{2+} efflux

Mechanism of action: **Directly inhibits Na^+/K^+ ATPase**, which decreases the trans-sarcolemmal Na^+ gradient, and therefore **indirectly inhibits $\text{Na}^+/\text{Ca}^{2+}$ exchanger**.

Use: **Systolic Heart Failure**



SMOOTH MUSCLE

LEARNING OBJECTIVES

SOM.MK.II.BPM1.2.MSK.2.PHYS.0072 Compare the similarities and differences in the mechanisms of actomyosin regulation in the smooth muscle vs striated muscle (cardiac and skeletal muscles)

SOM.MK.II.BPM1.2.MSK.2.PHYS.0073 Explain the concept of latch mechanism in the smooth muscle cells, and how these cells can develop and maintain force with a much lower rate of ATP hydrolysis than skeletal muscle

SOM.MK.II.BPM1.2.MSK.2.PHYS.0074 List the intracellular pathways that control the contraction and relaxation in smooth muscle cells and explain the differences between electromechanical and pharmacomechanical coupling

SOM.MK.II.BPM1.2.MSK.2.PHYS.0075 Compare the characteristics of the multi-unit and unitary smooth muscle types

SOM.MK.II.BPM1.2.MSK.2.PHYS.0076 Explain the myofilament calcium sensitization and desensitization mechanisms

Lecture Outline

- ❖ Location, functions, structures, characteristics
- ❖ Multi-Unit vs Unitary Smooth Muscle
- ❖ Smooth muscle contraction/relaxation
- ❖ Latch mechanism or state
- ❖ Ca^{++} sensitivity of the myofilaments

Smooth muscle location

Gastrointestinal tract: Propulsion of the food and control of sphincters

Cardiovascular: Regulation of blood flow and pressure

Renal: Regulation of renal blood flow and micturition (urination)

Reproductive: Contraction of uterus during labor & propulsion of sperm/oocyte

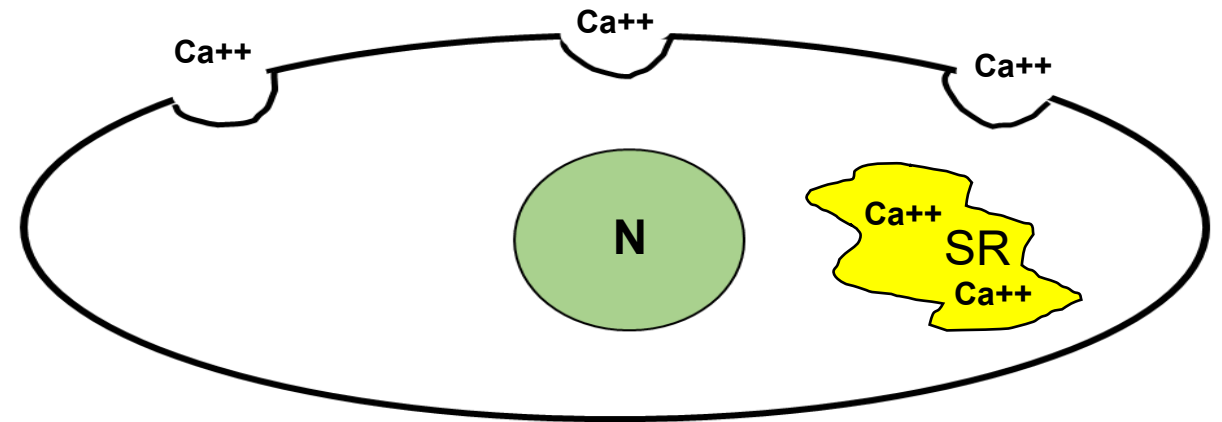
Respiratory tract: Regulation of bronchiolar diameter

Integument: Regulation of skin blood flow & arrector pili muscle

Eyes: Regulation of pupil size and curvature of lens

Smooth muscle cell structure

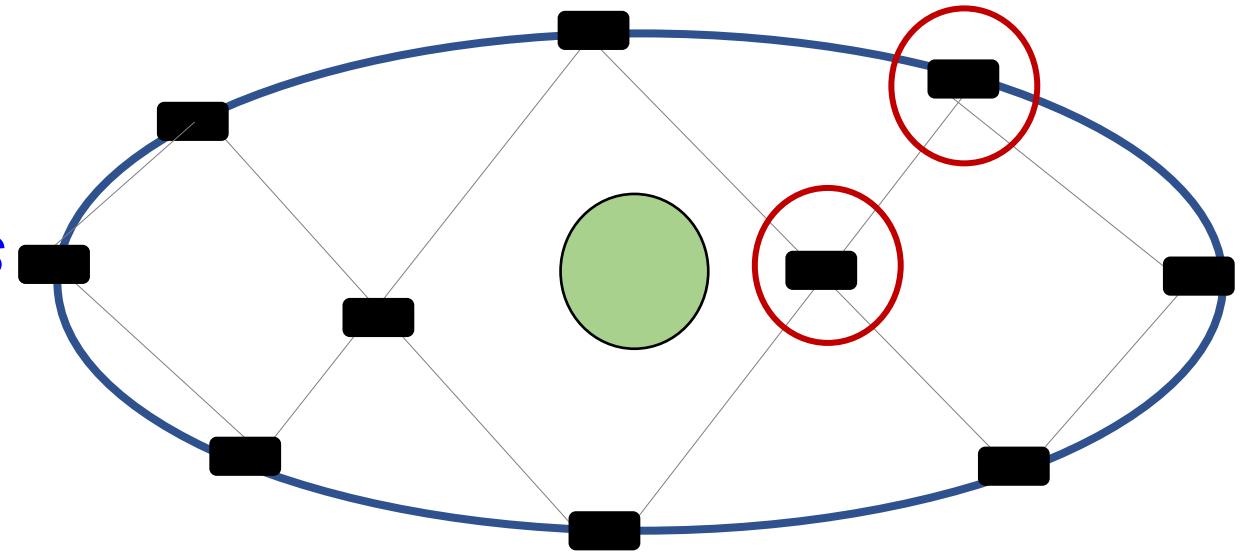
- Fusiform
- Diameter: 5-10 μm
- Length: 20-500 μm
- Non-striated pattern because **NO SARCOMERES**
- Caveolae: analogue to T-tubules
- Sarcoplasmic reticulum: poorly developed, but good Ca^{++} store for contraction.



N = Nucleus

Smooth muscle cell structure

- Dense bodies
(α -actinin; *analogous to Z-lines in striated muscles*)
- Membrane-associated dense bodies
- Intermediate filaments – make cytoskeleton

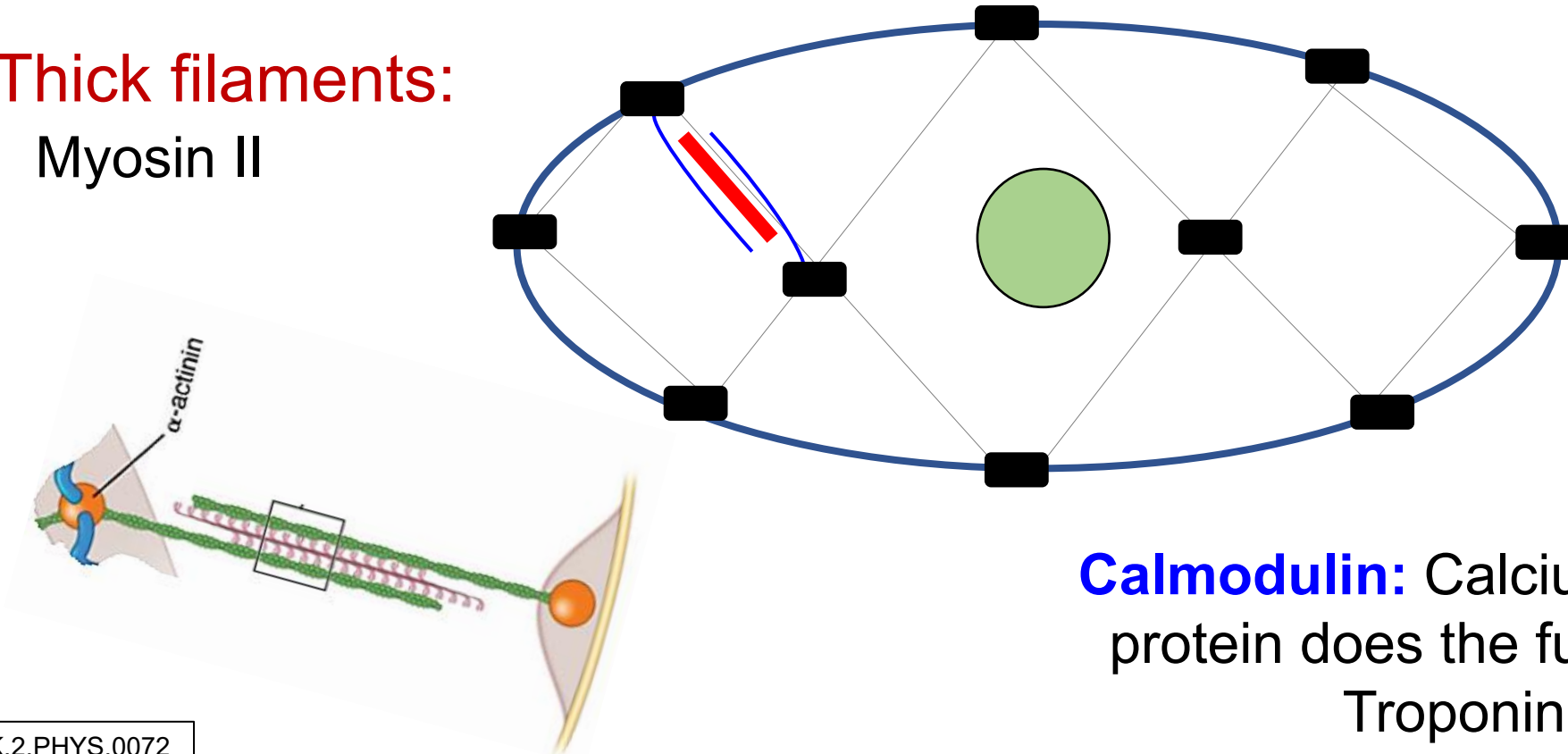


Contractile apparatus of smooth muscle

- Thin filaments:

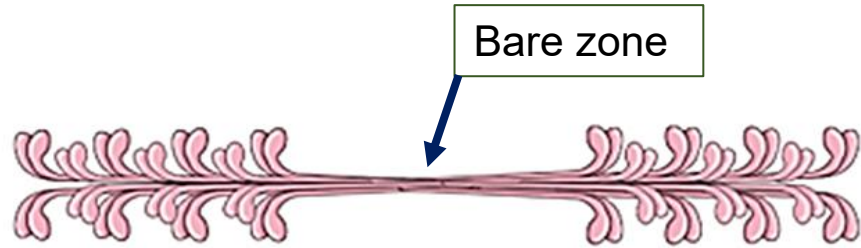
contains: Actin + Caldesmon + Calponin + Tropomyosin
Lack TROPONIN (regulatory protein in striated muscle)

- Thick filaments:
Myosin II

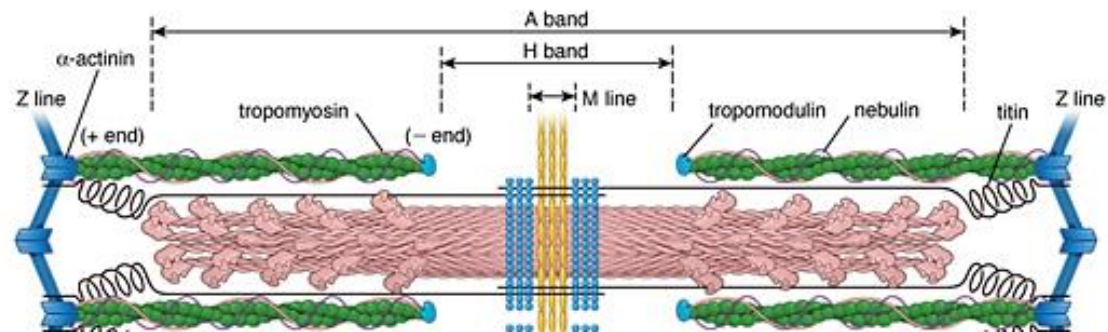


Calmodulin: Calcium binding protein does the function of Troponin

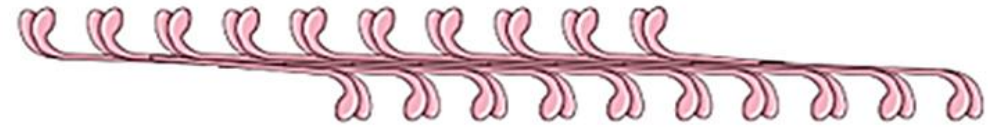
Striated muscle



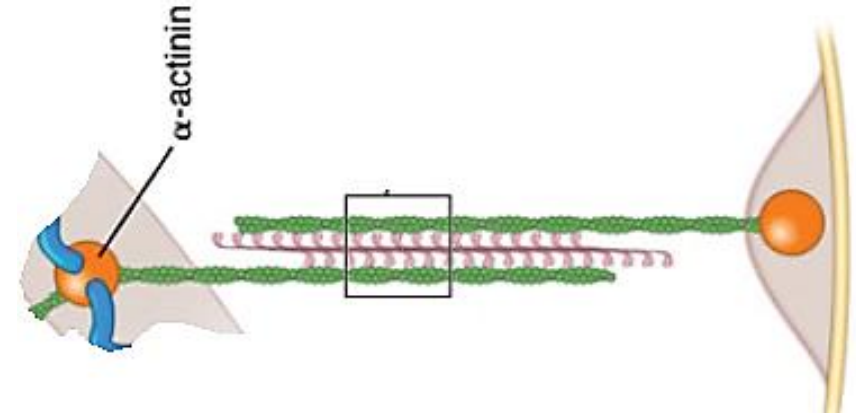
bipolar thick filament



Smooth muscle

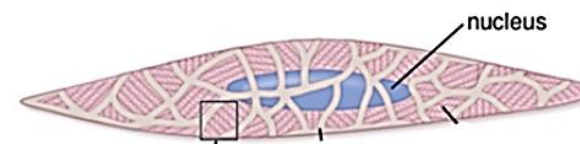


side-polar thick filament



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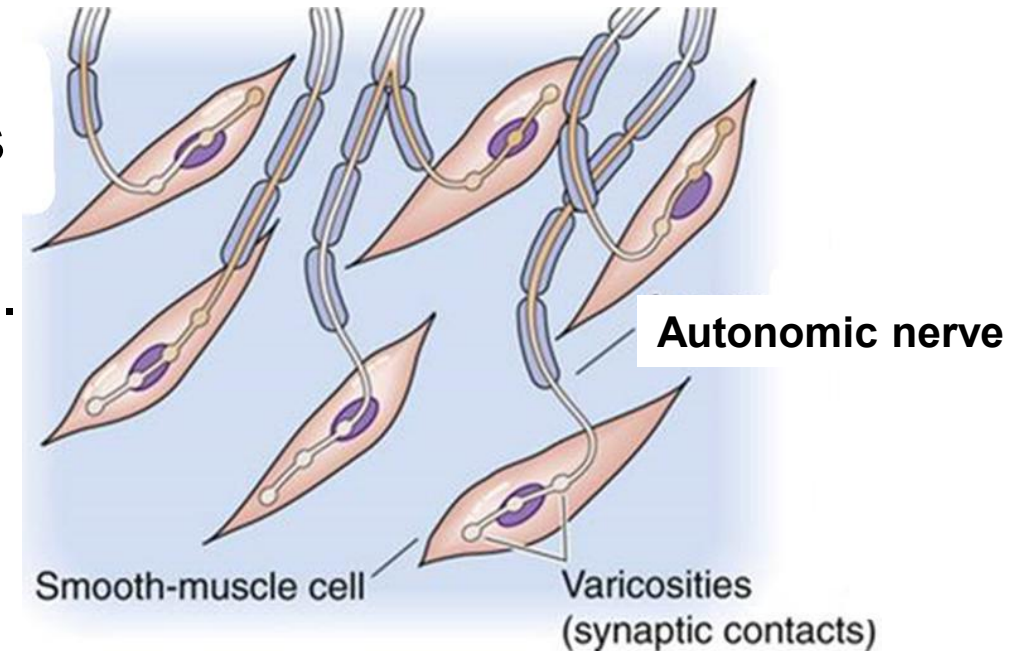
Comparison: Striated vs Smooth Muscle

Features	Striated muscle	Smooth muscle
Actomyosin regulation	Thin filament: Troponin C/ Tropomyosin	Thick filament: phosphorylation of the regulatory light chain of Myosin II (myosin-linked regulation) – Ca ⁺⁺ /CaM mediated
Contractile units	Sarcomere	No sarcomere
Thin filaments	Actin, Tropomyosin	
	Troponin	Calmodulin (No Troponin)
Thick filaments	Myosin II	
	Bipolar arrangement	Side-polar arrangement
	Higher amount of Myosin II	Lower amount of Myosin II
Thick to thin filament ratio	1:6	1:12

Multi-Unit vs Unitary Smooth Muscle

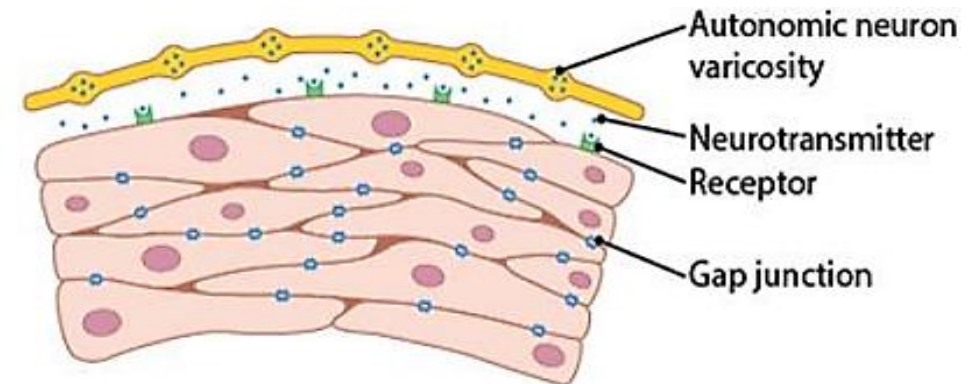
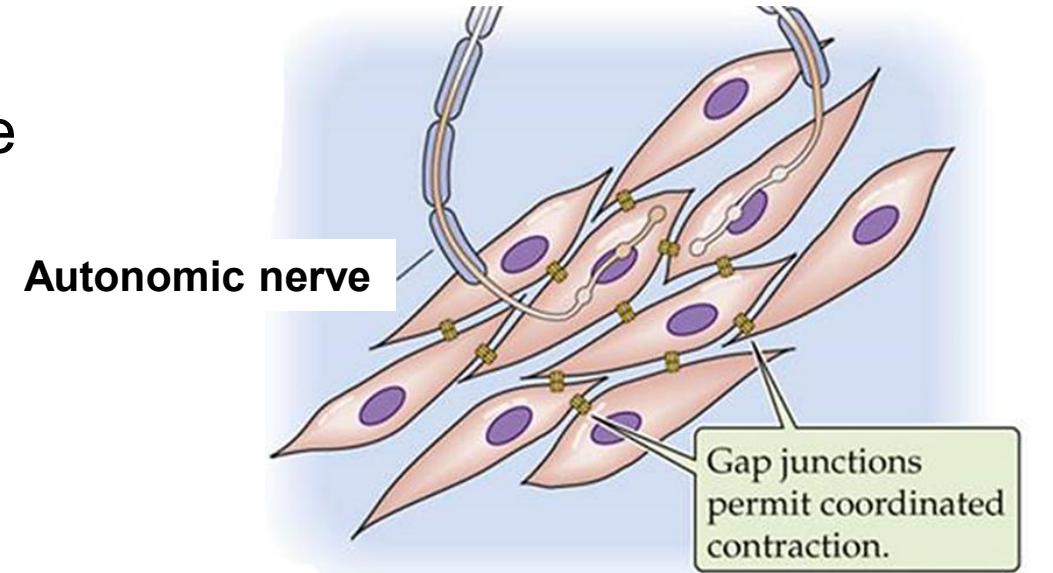
Multi-Unit Smooth Muscles

- Each fiber is innervated by a **single nerve ending**
- Every fiber can contract independently and is controlled by a nerve signal
- A finer motor control on the muscles possible.
- Found in:
 - Ciliary muscle of the eye,
 - Sphincter pupillae muscle,
 - Dilator pupillae muscle,
 - Arrector pili muscles.



Unitary (or Single-unit) Smooth Muscles

- Many muscle fibers are innervated by one neuron
- Electrically coupled by GAP junctions
- Contract together (syncytium)
- Found in:
Walls of most of the internal organs (viscera) of the body like:
Gastrointestinal tract,
Bile ducts,
Ureters,
Uterus,
Blood vessels.

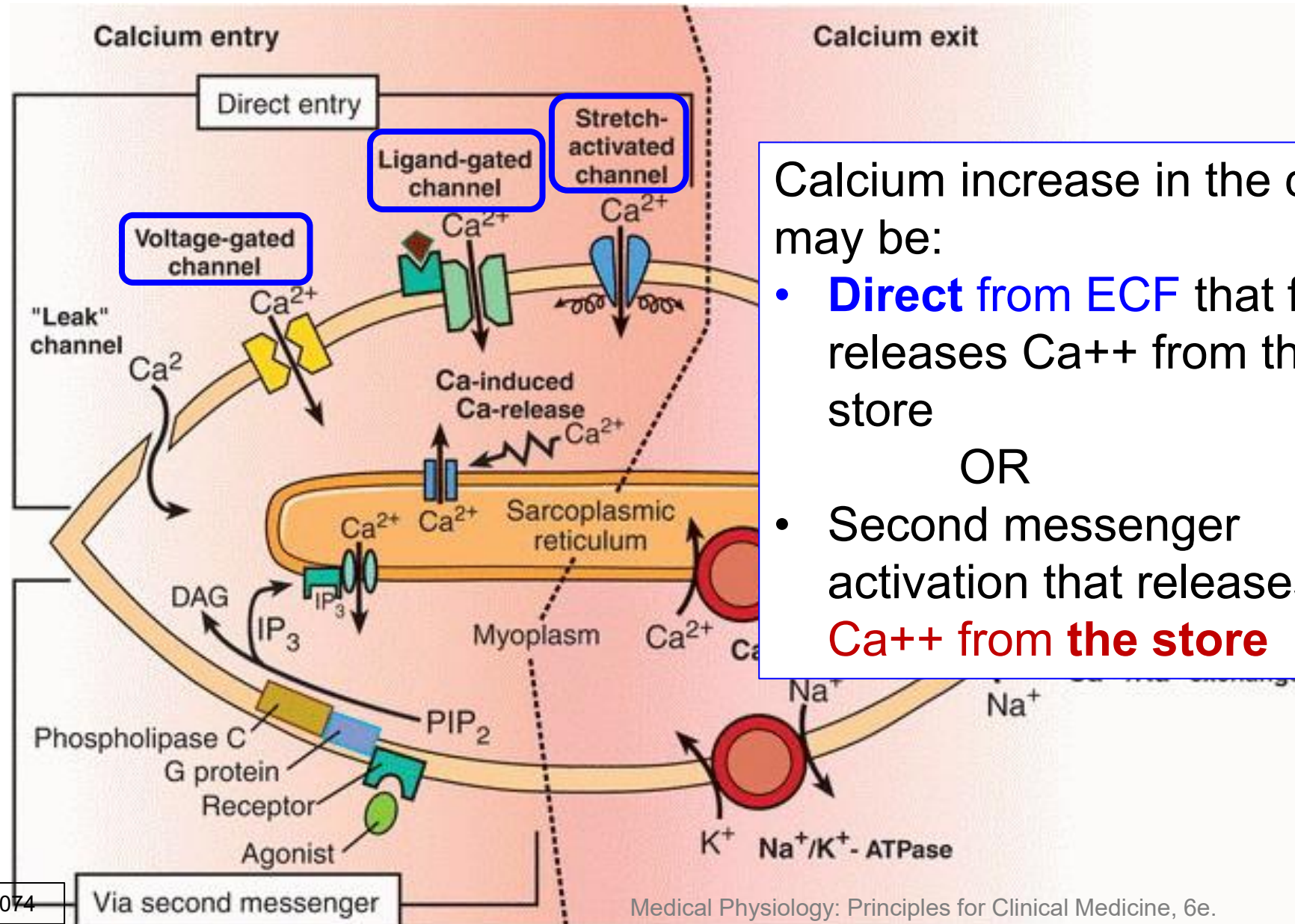


Smooth muscle contraction/relaxation

Source of Calcium initiating Smooth muscle contraction

**Direct
from ECF**

**Second
messenger
mediated**



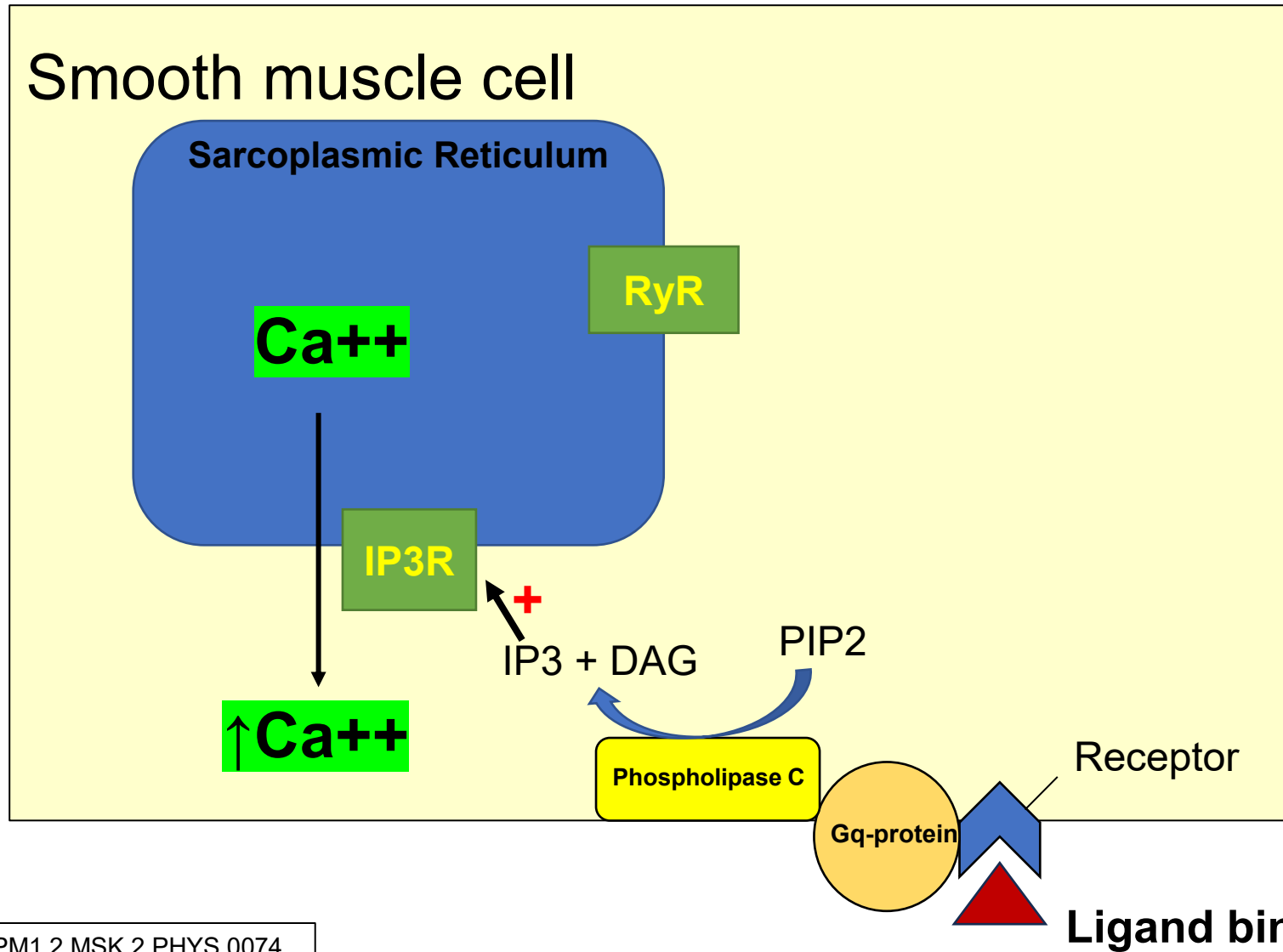
Calcium increase in the cell may be:

- **Direct** from ECF that further releases Ca⁺⁺ from the store

OR

- Second messenger activation that releases **Ca⁺⁺ from the store**

Calcium release from store: Gq-protein-PLC-IP3 pathway activation



EXAMPLES

- Norepinephrine, Epinephrine & Dopamine (via alpha adrenoreceptors) in vascular sm muscles
- Acetylcholine (via M receptors) in GIT
- Angiotensin II (via angiotensin II receptors) in vascular sm muscles
- Endothelin (via ET_A receptors)

Contraction of smooth muscle occurs by TWO ways: **with** or **without** an action potential

Action Potential through nerve stimulation



Activation of a voltage-gated calcium channel



Increase in intracellular Ca^{+2}



This type of initiation of contraction is called,
“**Electromechanical coupling**”.

Binding of a **chemical substance (ligand)** to membrane receptor



Activation of membrane channel or activation of Gq-protein



Increase in intracellular Ca^{+2}



This type of initiation of contraction is called,
“**Pharmacomechanical coupling**”.

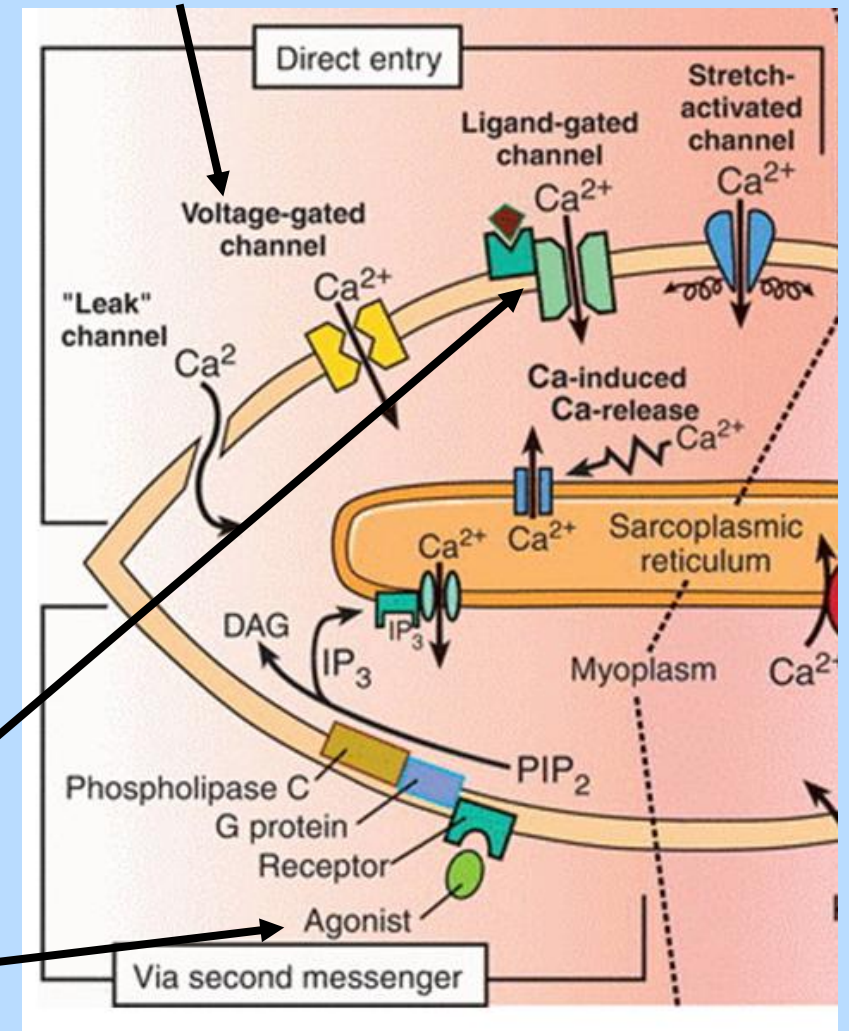
Electromechanical versus Pharmacomechanical coupling

Electromechanical coupling produces a **phasic** smooth muscle contraction that is short-lived. Voltage-gated Ca^{++} channels involved. E.g., Bladder contraction by pelvic nerve stimulation during micturition.

Pharmacomechanical coupling produces a **tonic smooth muscle contraction** that is prolonged and suitable for functioning of the unitary smooth muscles. E.g., Binding of Hormones, Neurotransmitters and drugs cause contraction. This is the **Predominant mechanism** of smooth muscle contraction.

Action potential

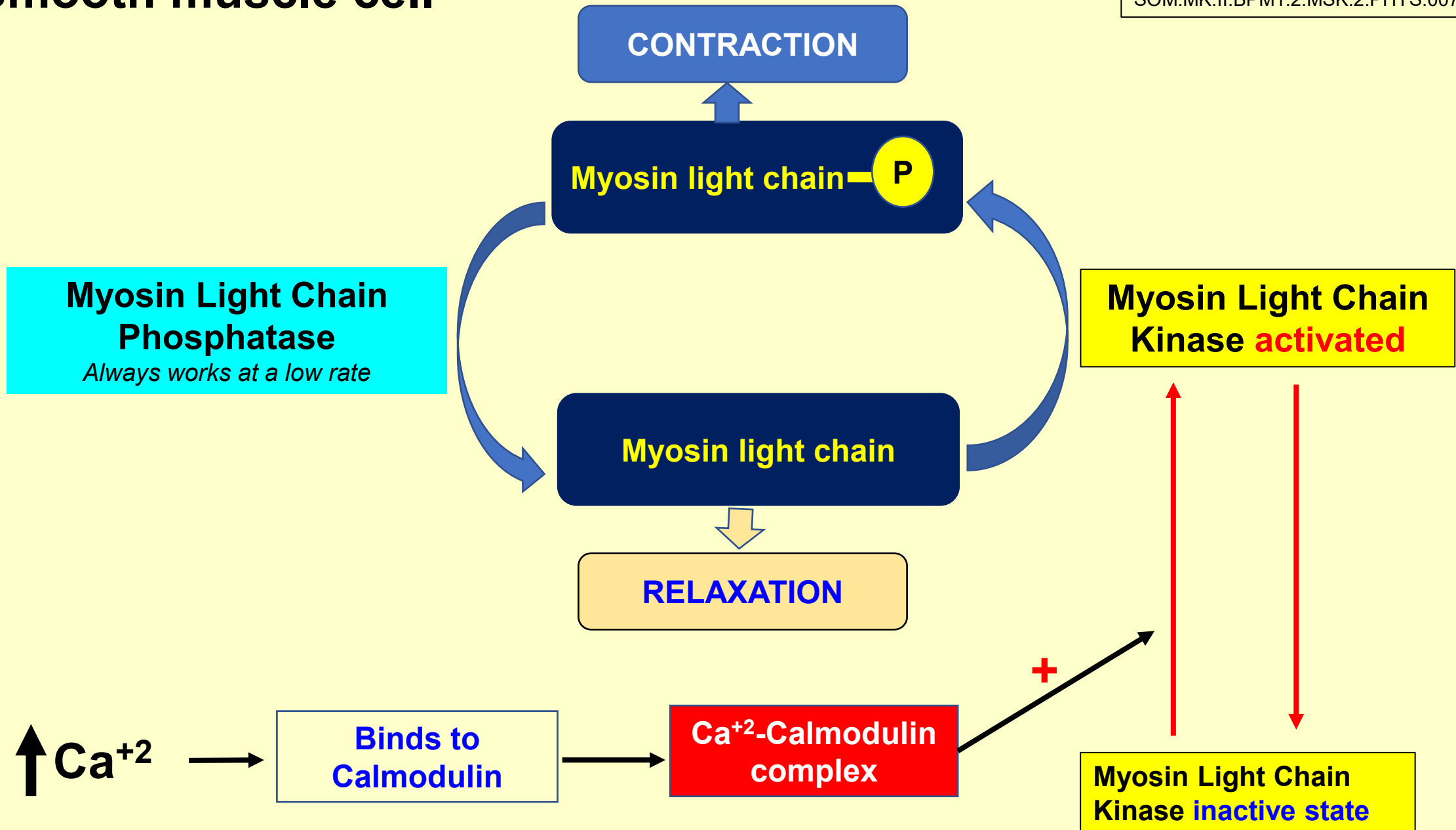
Ligand



Molecular Mechanism of Smooth muscle contraction/relaxation

Smooth muscle cell

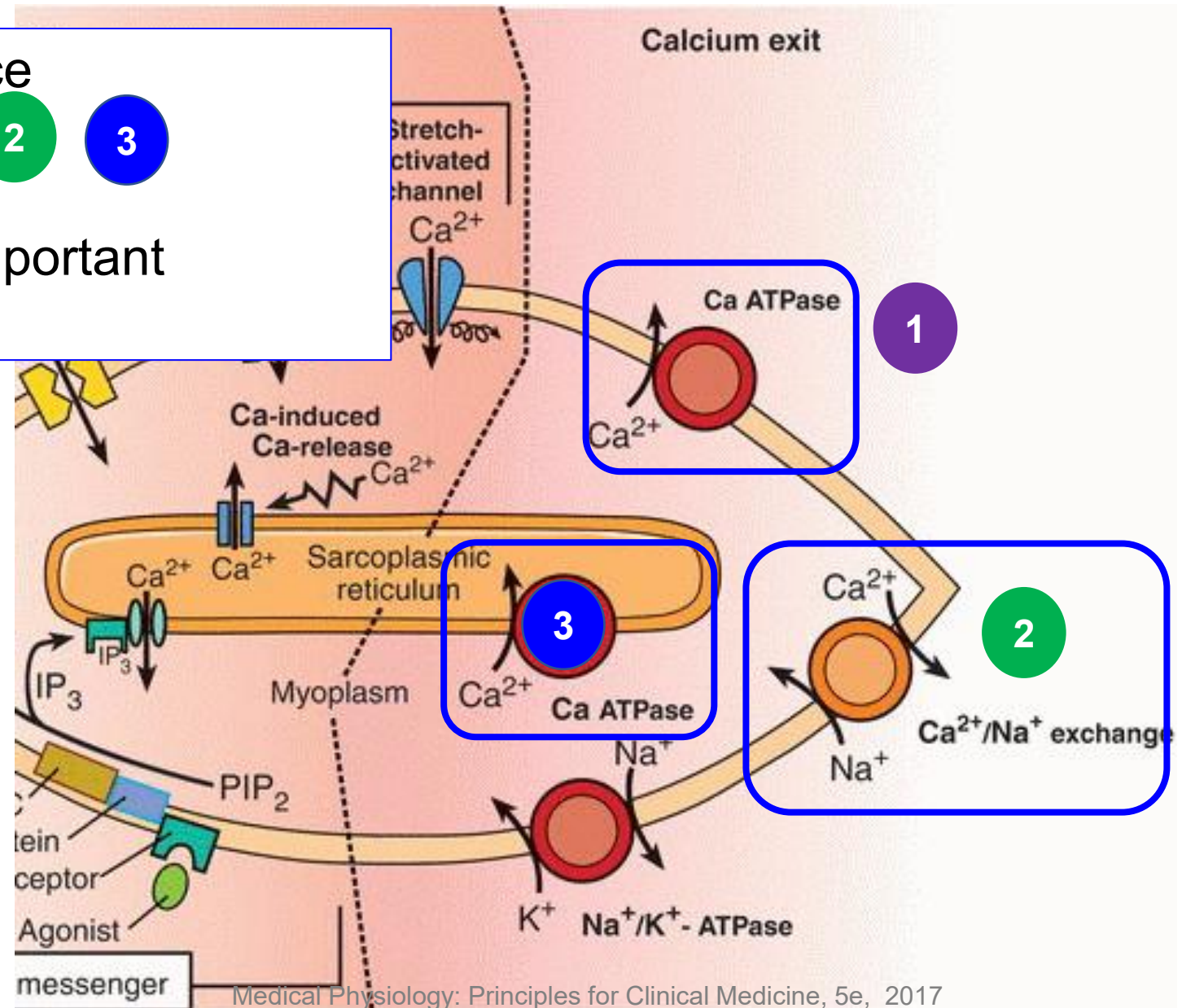
SOM.MK.II.BPM1.2.MSK.2.PHYS.0074



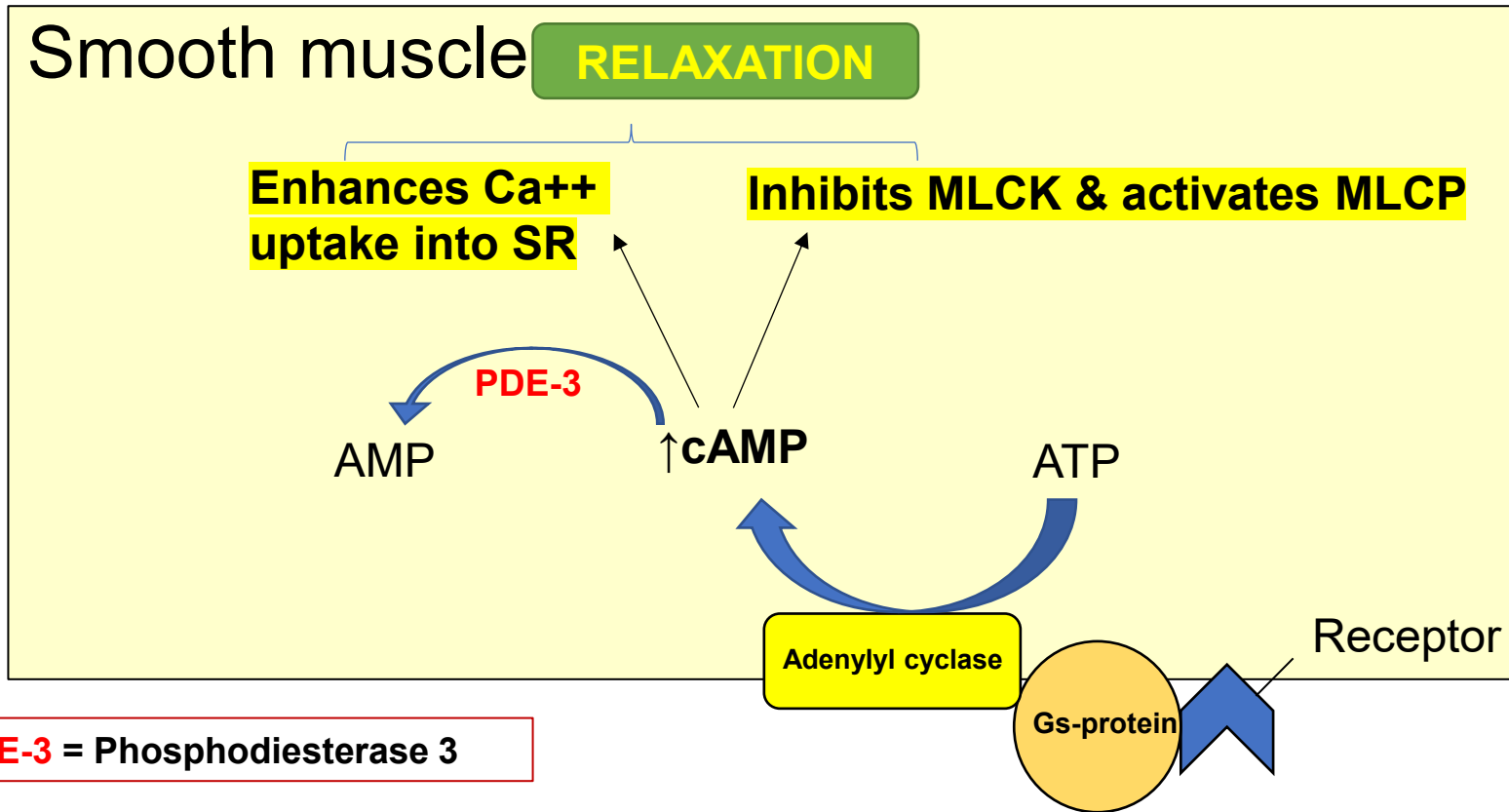
Calcium exit: RELAXATION of smooth muscles

- Three mechanisms that reduce intracytoplasmic Ca^{+2} : 1 2 3
- Mechanism 3 is the most important

Decrease in Ca^{+2}
↓
Decreases **MLCK** activity &
increases MLCP activity
↓
Muscle relaxation



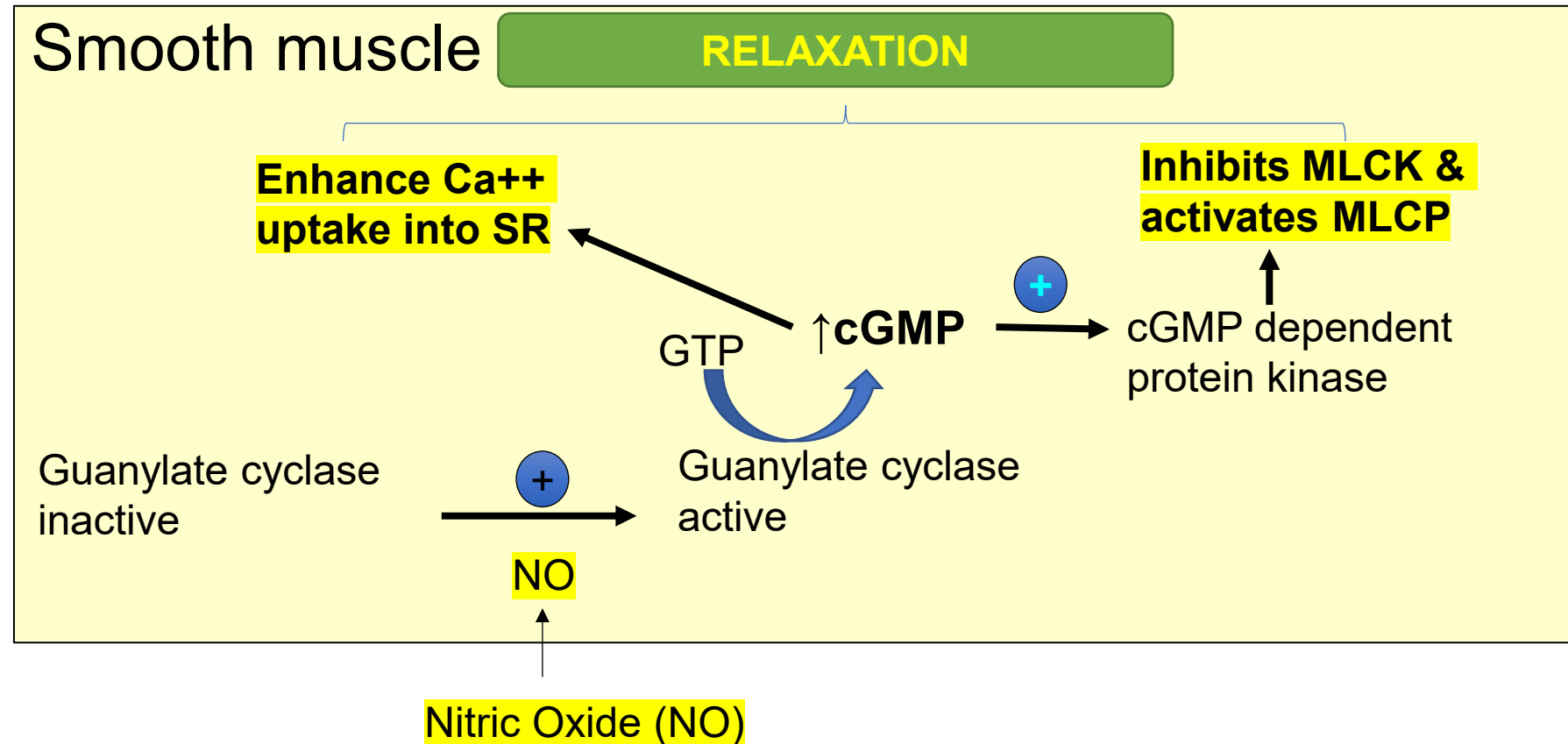
Increased cAMP Relaxes Smooth muscles: Examples



- Norepinephrine, Epinephrine (via beta2 adrenoreceptor) in bronchial & uterine smooth muscles
- Dopamine (via D1 receptors) in renal vascular muscles
- Histamine (via H2 receptors) in skin blood vessels

- Beta-2 agonist, Albuterol inhaler is used in Asthma for bronchodilation
- Terbutaline, beta-2 agonist used to relax uterine smooth muscles in pregnant women with premature labor

Nitric Oxide Relaxes Smooth muscles: cGMP pathway



- **Nitrates** used in angina increase NO in vascular smooth muscles improve blood flow to heart.
- Drugs that increase cGMP in vascular smooth muscles cause vasodilation and reduce blood pressure in severe hypertension (Examples: **Hydralazine, Nitroprusside**)

**Latch mechanism or state and
Calcium sensitivity of the myofilaments**

Latch state and its importance in smooth muscle

Definition: Smooth muscles can maintain a high level of force of contraction with **a very low** level of calcium and ATP consumption.

Cross-bridge cycling occurs at a **slow rate** so that **each remains in the attached state for a longer time with least energy and calcium.**

Latch is NOT a rigor state.

Importance: This is helpful in organs where a sustained contraction for a long period is required *with **least energy expenditure***. Example: blood vessels to maintain BP for the entire life.

Outline

Location, functions, structures, characteristics

Multi-Unit vs Unitary Smooth Muscle

Smooth muscle contraction/relaxation

Latch mechanism or state

Ca⁺⁺ sensitivity of the myofilaments

Calcium sensitization/de-sensitization of smooth muscles

- MLCK and MLCP activities alter calcium requirements of contraction.
- **Increased MLCK** activity and **decreased MLCP** activity: Can produce a stronger contraction even with a lower-than-normal calcium. This is called **calcium sensitization**.
- **Decreased MLCK** activity and **increased MLCP** activity: Can produce a weaker contraction even with a higher-than-normal calcium. This is called **calcium de-sensitization**.

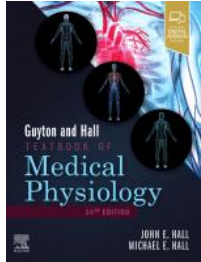
Distinguishing Characteristics of 3 different Muscle Types

Features	Skeletal Muscle (Striated)	Cardiac Muscle (Striated)	Smooth Muscle (Nonstriated)
Cells	Thick, long, unbranched, cylindric	Branched, cylindric	Small, spindle-shaped
Nuclei per cell	Many; peripheral	One or two; central	One; central
Filament ratio	6 thin: 1 thick	6 thin: 1 thick	12 thin: 1 thick
Sarcoplasmic reticulum and myofibrils	Highly organized sarcoplasmic reticulum surrounds distinct myofibrils	Less organized sarcoplasmic reticulum; irregular myofibrils	Poorly organized sarcoplasmic reticulum; no myofibrils
T tubules	At A-I band junctions; form triads	At Z lines; form diads	None
Motor end-plates	Present	Absent	Absent
Motor control	Voluntary	Involuntary	Involuntary
Other	Prominent fascicles, thick perimysium and epimysium	Intercalated disks at cell-to-cell junctions	Abundant caveolae, cells overlap

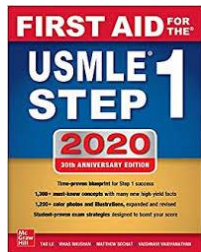
Distinguishing Characteristics of 3 different Muscle Types.

Property	Skeletal	Cardiac	Smooth
Ca ²⁺ sensor	Troponin C	Troponin C	Calmodulin
Excitation	neuromuscular	Pacemaker potentials, electrical coupling	Synaptic, hormone, electrical coupling, pacemaker potentials
Excitation contraction coupling	L-Type (DHP receptor) voltage sensor Store release Ryr	L-Type (DHP) calcium channel Store release Ryr	Voltage gated Ca ²⁺ channels, Ca ²⁺ and IP ₃ mediated Ca ²⁺ stores, store operated Ca ²⁺ channels
Contraction termination	ACh breakdown	Myocyte repolarization	Ratio of MLCK to MLCP activity
Twitch duration	20-200 msec	200-400 msec	200 – sustained latch
Force regulation	Frequency and fiber recruitment	Ca ²⁺ concentration	Balance of MLCK – MLCP MLCK regulation by PKA, PKC, Ca ²⁺ -Cal Latch state
Electrical Activity	Spikes	Spike and plateau	Spike, plateau, graded potentials, slow waves
T tubules	Yes	Yes	No
Gap junctions	No	yes	Multiunit – few Unitary - many

Other recommended readings



Guyton and Hall Textbook of Medical Physiology, Fourteenth Edition, 2021. Chapter 8: Excitation and Contraction of Smooth Muscle and Chapter 9: Physiology of the cardiac muscle.



Le, Tao and Bhushan, Vikas. First Aid for the USMLE Step 1 2021, Thirty First edition. New York: McGraw-Hill Education, 2021.

Any Questions ?

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